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Asymptotic model-based estimation of a wake oscillator for a tethered sphere in uniform flow

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ABSTRACT

We derive and investigate the nonlinear dynamics of a tethered sphere in uniform flow. A Lagrangian based model augmented by wake oscillators for the streamwise vertical and transverse directions enables model-based estimation of both structural and aeroelastic parameters via asymptotic analysis of internal resonance conditions between the transverse wake frequency and its structural counterpart. Validation of the proposed methodology is demonstrated by comparison of results with those of a benchmark experiment for a light sphere in water conducted by Govardhan and Williamson (1997) and by analysis of a spherical aerostat experiment conducted by Coulomb-Pontbriand and Nahon (2009). A numerical bifurcation analysis of the validated system reveals existence of possible quasiperiodic and non-stationary solutions that are consistent with documented instabilities of aerostats in severe environmental conditions and shed light on control mechanisms required for suppression of finite amplitude limit-cycle oscillations.

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1. Introduction

Tethered aerostats are constrained lighter-than-air systems which have been widely used for remote sensing purposes for over a century. Recently, tethered aerostats have been proposed for capturing wind (Vermillion et al., 2014) and solar energy (Aglietti et al., 2008). Additional examples of tethered balloon systems have been proposed for calibration of structural parameters of a system designed for Venusian atmosphere exploration (Hall et al., 2008) and for geoengineering purposes (Davidson et al., 2012).

In light of the wide range of applications, there have been efforts to analyze the aerostat system stability (DeLaurier, 1972, 1981; Dietl et al., 2011) and to estimate its aerodynamic forces (Jones and DeLaurier, 1983). Theoretical lumped-mass models have been primarily employed for system response prediction (Coulomb-Pontbriand and Nahon, 2009; Rajani et al., 2010; Redi et al., 2011; Kang and Lee, 2009) or control design (Nahon et al., 2002), whereas continuum based models were also developed (Kim and Perkins, 2002; Stanney and Rahn, 2006) to capture the multimode response of cable dynamics. Related numerical (Behara et al., 2011; Lee et al., 2013) and experimental (Lee et al., 2013; Williamson and Govardhan, 2004; Govardhan and Williamson, 2005; van Hout et al., 2010) research that highlight the abundance of flow-induced instabilities of tethered spheres in uniform flow reveal a complex bifurcation structure that includes different types of coexisting periodic responses (Govardhan and Williamson, 1997) and intermittent or non-stationary structural dynamics (Jauvtis et al., 2001; Eshbal et al., 1997). Similar behavior was also observed in a small-scale field experiment of a spherical aerostat subject to non-stationary wind conditions (Coulomb-Pontbriand and Nahon, 2009).

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To date, there is no documentation of a comprehensive and validated theoretical model that enables prediction of the onset of vortex-induced vibration (VIV) for tethered spheres or aerostats. Thus, the objective of this research is to derive and investigate a phenomenological low-order dynamical system model for a tethered spherical aerostat that still captures the VIV stability threshold for the onset of self-excited limit-cycle oscillation.

This paper is organized as follows. Following an introduction in Section 1, we formulate in Section 2 a nonlinear dynamical system including a Lagrangian based derivation of the tethered spherical aerostat augmented by a nonlinear wake oscillator model, which incorporates documented mechanisms required to predict the VIV threshold. In Section 3, we conduct an equilibrium analysis to determine conditions for the unknown linear system parameters. In Section 4, we use the asymptotic multiple-scales method to estimate the unknown nonlinear system parameters. The resulting system parameters are validated by comparison of system response with the transverse limit-cycle amplitudes obtained from a benchmark experiment (Govardhan and Williamson, 1997). In Section 5, we numerically investigate the proposed model tailored to describe conditions of a small scale spherical aerostat model (Coulomb-Pontbriand and Nahon, 2009). We conclude with some closing remarks in Section 6.

2. Problem formulation

We derive the equations of motion for a strongly nonlinear tethered sphere (Fig. 1) using a Lagrangian approach (Meirovitch, 1988). The dynamical system is completed by a wake oscillator model that incorporates both a van der Pol term (Hartlen and Currie, 1970; Blevins, 2001) to ensure self-excited limit-cycle oscillations (LCO) beyond the flutter (or Hopf) threshold and quadratic interaction terms proposed and validated by Kim and Perkins (2002) for VIV of cable dynamics. The latter was recently used to describe the nonlinear VIV of a spherical mathematical pendulum in uniform flow (Shoshani and Gottlieb, 2011).

2.1. The elastic pendulum

Following Gottlieb (1997), the kinetic and potential energies of a tethered aerostat system defined for a reference frame located at the static equilibrium position (see Fig. 1) are

$$\begin{aligned} \mathrm{KE} &= \frac{1}{2}m\left(\dot{X}^2+\dot{Y}^2+\dot{Z}^2\right)\\ \mathrm{PE} &= mgZ+\frac{1}{2}k(\sqrt{X^2+Y^2+(l+Z)^2}-l_0)^2, \end{aligned} \tag{1}$$

where g is acceleration of gravity, k is elastic spring stiffness, l_0 is spring free length and l is the equilibrium length defined by $l=l_0+(\rho_F V-m)g/k$. The equations of motion for the tethered system are derived from the Lagrangian function $\mathcal{L}=\mathrm{KE}-\mathrm{PE}$:

$$m\begin{pmatrix}\ddot{X}\\ \ddot{Y}\\ \ddot{Z}\end{pmatrix}+E\begin{pmatrix}X\\ Y\\ l+Z\end{pmatrix}+\begin{pmatrix}0\\ 0\\ mg\end{pmatrix}=\begin{pmatrix}Q_X\\ Q_Y\\ Q_Z\end{pmatrix}, \tag{2}$$

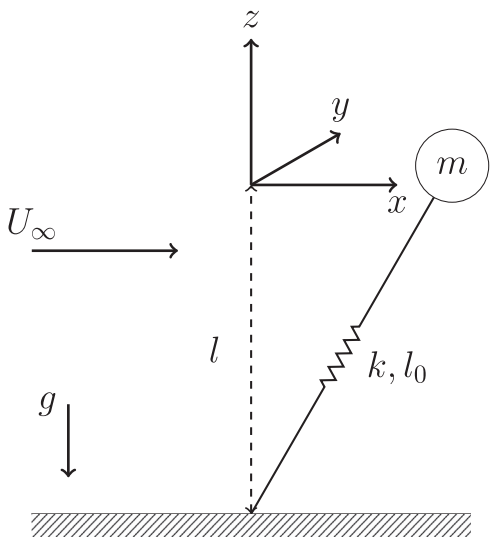


Fig. 1. Definition sketch of a tethered sphere.

where $E=k(1-l_0/\sqrt{X^2+Y^2+(l+Z)^2})$. The generalized forces Q_X, Q_Y and Q_Z induced by the hydrodynamic effects of buoyancy, added mass, drag, lift and equivalent linear structural damping are

$$\begin{pmatrix} Q_X \\ Q_Y \\ Q_Z \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \\ \rho_F V g \end{pmatrix} - \rho_F V C_A \begin{pmatrix} \ddot{X} \\ \ddot{Y} \\ \ddot{Z} \end{pmatrix} + \frac{1}{2}\rho_F A_S U_\infty^2 \begin{pmatrix} C_D \\ C_T \\ C_L \end{pmatrix} - \delta \begin{pmatrix} \dot{X} \\ \dot{Y} \\ \dot{Z} \end{pmatrix}. \tag{3}$$

We note that for a sphere the added mass coefficient is $C_A=0.5$ (Newman, 1977) and the aerodynamic coefficients for drag (C_D), transverse (C_T) and lift (C_L) directions are time dependent variables to be defined by the wake oscillator (Blevins, 2001). By merging terms, the dimensional equation of motion can be written as

$$(m+\rho_F V C_A)\begin{pmatrix} \ddot{X} \\ \ddot{Y} \\ \ddot{Z} \end{pmatrix} + E\begin{pmatrix} X \\ Y \\ l+Z \end{pmatrix} = \frac{1}{2}\rho_F A_S U_\infty^2 \begin{pmatrix} C_D \\ C_T \\ C_L \end{pmatrix} - \delta \begin{pmatrix} \dot{X} \\ \dot{Y} \\ \dot{Z} \end{pmatrix} + \begin{pmatrix} 0 \\ 0 \\ (\rho_F V - m)g \end{pmatrix}. \tag{4}$$

2.2. The wake oscillator

Following Kim and Perkins (2002) we define the drag C_D, transverse C_T and lift C_L coefficients as functions of mean and fluctuating terms:

$$C_D = \beta_D + p(\tau), \quad C_T = q(\tau), \quad C_L = \beta_L + r(\tau), \tag{5}$$

where β_D and β_L are the nondimensional mean drag and lift respectively, and $p(\tau), q(\tau), r(\tau)$ are the respective time dependent variables for drag, transverse and lift. Following Hartlen and Currie (1970) we incorporate a van der Pol oscillator model for the wake augmented by linear and quadratic coupling terms motivated by Kim and Perkins (2002) and recently used for a spherical pendulum with a constant length tether by Shoshani and Gottlieb (2011):

$$\begin{aligned} &p_{\tau\tau} + \omega_p^2 p = (\alpha_p - \gamma_p p^2)p_\tau - (k_{p2} q q_\tau + k_{p3} q_\tau^2) + \mu_p x_\tau \\ &q_{\tau\tau} + \omega_q^2 q = (\alpha_q - \gamma_q q^2)q_\tau - (k_{q2} p q_\tau + k_{q3} p_\tau q + k_{q4} p_\tau q_\tau) \\ &\qquad\qquad - (k_{q6} r q_\tau + k_{q7} r_\tau q + k_{q8} r_\tau q_\tau) + \mu_q y_\tau \\ &r_{\tau\tau} + \omega_r^2 r = (\alpha_r - \gamma_r r^2)r_\tau - (k_{r2} q q_\tau + k_{r3} q_\tau^2) + \mu_r z_\tau. \end{aligned} \tag{6}$$

We note that Kim and Perkins (2002) proposed additional quadratic terms $k_1 pq$ and $k_5\, q^2$. However, in order to ensure a unique zero equilibrium for the wake oscillator, we neglect these coefficients.

2.3. The nondimensional equations for the elastic pendulum

We rescale time ($\tau=\omega_s t$) by the natural frequency defined by the tether stiffness and combined body added mass: $\omega_s=\sqrt{k/(m+\rho_F V C_A)}$, and we rescale the generalized coordinates ($x=X/l, y=Y/l, z=Z/l$) by the equilibrium length. Therefore the complete model for the system is the following:

$$\begin{pmatrix} x_{\tau\tau} \\ y_{\tau\tau} \\ z_{\tau\tau} \end{pmatrix} + R\begin{pmatrix} x \\ y \\ 1+z \end{pmatrix} = \eta u^2 \begin{pmatrix} \beta_D + p \\ q \\ \beta_L + r \end{pmatrix} - \delta \begin{pmatrix} x_\tau \\ y_\tau \\ z_\tau \end{pmatrix} + \begin{pmatrix} 0 \\ 0 \\ 1-\alpha \end{pmatrix}$$

$$\begin{aligned} &p_{\tau\tau} + \omega_p^2 p = (\alpha_p - \gamma_p p^2)p_\tau - (k_{p2} q q_\tau + k_{p3} q_\tau^2) + \mu_p x_\tau \\ &q_{\tau\tau} + \omega_q^2 q = (\alpha_q - \gamma_q q^2)q_\tau - (k_{q2} p q_\tau + k_{q3} p_\tau q + k_{q4} p_\tau q_\tau) \\ &\qquad\qquad - (k_{q6} r q_\tau + k_{q7} r_\tau q + k_{q8} r_\tau q_\tau) + \mu_q y_\tau \\ &r_{\tau\tau} + \omega_r^2 r = (\alpha_r - \gamma_r r^2)r_\tau - (k_{r2} q q_\tau + k_{r3} q_\tau^2) + \mu_r z_\tau, \end{aligned} \tag{7}$$

where $R=1-\alpha/\sqrt{x^2+y^2+(1+z)^2}$ and the nondimensional coefficients are

$$\alpha = \frac{l_0}{l}, \quad \delta = \frac{\omega_s \delta}{k}, \quad \eta = \frac{3}{4L^*(M^*+C_A)}, \quad M^* = \frac{m}{\rho_F V}, \quad L^* = \frac{l}{d}, \quad u = \frac{U_\infty}{\omega_s d}. \tag{8}$$

Note that α defines the tether pretension, δ represents linear structural damping, η is an inverse added mass coefficient, M^* is the sphere mass ratio ($M^*<1$ defines a light sphere), L^* is the normalized tether length and u is a normalized flow velocity. The relation between the normalized flow velocity u and the standard reduced velocity ($U^*=U_\infty/f_n d$; f_n a natural frequency of the tethered system) is $u=U^* f_n/\omega_s$.

2.4. The reduced-order model

To simplify the stability analysis of the equilibrium, we assume that motion about the mean drag and lift are sufficiently small in comparison with the transverse component to yield a reduced-order model which couples the transverse displacement y with its corresponding wake q:

$$\begin{aligned} &y_{\tau\tau}+\overline{R}y=\eta u^2 q-\delta y_\tau \\ &q_{\tau\tau}+\omega_q^2 q=(\alpha_q-\gamma_q q^2)q_\tau+\mu_q y_\tau, \end{aligned} \tag{9}$$

where $\overline{R}=1-\alpha/\sqrt{x^{*2}+y^2+(1+z^*)^2}$ and x^* and z^* are the equilibrium position of x and z components that can be estimated from the mean of given experimental measurements. Note that the restoring force $\overline{R}y$ in Eq. (9) is antisymmetric with respect to y.

3. Equilibrium analysis

The objective of the analysis in this section is to employ a model-based estimation methodology to evaluate the unknown linear system parameters, given sufficient data from the system equilibrium position. Specifically, in Subsection 3.1 we determine the structural pretension α via analysis of the full system equilibrium and in Subsection 3.2 we determine the linear wake and interaction coupling coefficients α_q and μ_q via a local stability analysis of the reduced-order system at the Hopf threshold for VIV.

3.1. The full system equilibrium

To find the equilibrium of the system in Eq. (7), we set all the time derivative terms to zero and verify that the wake oscillator equilibrium is zero ($p^*=q^*=r^*=0$). Substitution of the zero wake equilibrium into the structural equations yields the following equilibrium positions for the sphere:

$$\begin{aligned} x^* &= \left(1+\frac{\alpha}{\sqrt{(\eta u^2\beta_D)^2+(\eta u^2\beta_L+1-\alpha)^2}}\right)\eta u^2\beta_D \\ y^* &= 0 \\ z^* &= \left(1+\frac{\alpha}{\sqrt{(\eta u^2\beta_D)^2+(\eta u^2\beta_L+1-\alpha)^2}}\right)(\eta u^2\beta_L+1-\alpha)-1. \end{aligned} \tag{10}$$

Note that there is a simple relation between the streamwise x^* and vertical z^* positions at equilibrium:

$$\frac{1+z^*}{x^*}=\frac{\beta_L}{\beta_D}+\frac{1-\alpha}{\eta u^2\beta_D}. \tag{11}$$

We linearize the dynamical system in Eq. (7) about the equilibrium in Eq. (10) (and $p^*=q^*=r^*=0$) to yield the standard matrix form as follows:

$$\boldsymbol{x}_{\tau\tau}+\boldsymbol{C}x_\tau+\boldsymbol{K}x=0, \tag{12}$$

where

$$\boldsymbol{C}=\begin{pmatrix} \delta & 0 & 0 & 0 & 0 & 0 \\ 0 & \delta & 0 & 0 & 0 & 0 \\ 0 & 0 & \delta & 0 & 0 & 0 \\ -\mu_p & 0 & 0 & -\alpha_p & 0 & 0 \\ 0 & -\mu_q & 0 & 0 & -\alpha_p & 0 \\ 0 & 0 & -\mu_p & 0 & 0 & -\alpha_p \end{pmatrix}$$

$$\boldsymbol{K}=\begin{pmatrix} a_{21} & 0 & a_{25} & \eta u^2 & 0 & 0 \\ 0 & R^* & 0 & 0 & \eta u^2 & 0 \\ a_{25} & 0 & a_{65} & 0 & 0 & \eta u^2 \\ 0 & 0 & 0 & \omega_p^2 & 0 & 0 \\ 0 & 0 & 0 & 0 & \omega_q^2 & 0 \\ 0 & 0 & 0 & 0 & 0 & \omega_r \end{pmatrix}, \tag{13}$$

$R^* = 1-\alpha[x^{*2}+(1+z^*)^2]^{-1/2}$ is the equilibrium value for R in Eq. (7) and

$$a_{21} = 1-\frac{\alpha(1+z^*)^2}{[x^{*2}+(1+z^*)^2]^{3/2}}, a_{25} = \frac{\alpha x^*(1+z^*)}{[x^{*2}+(1+z^*)^2]^{3/2}}, a_{65} = 1-\frac{\alpha x^{*2}}{[x^{*2}+(1+z^*)^2]^{3/2}}. \tag{14}$$

Note that the x and z equations above are coupled in both stiffness $\boldsymbol{K}$ and damping $\boldsymbol{C}$. The undamped natural frequencies are thus computed by equating the determinant of $(\boldsymbol{K}-\omega^2\boldsymbol{I})$ to zero to yield the non-dimensional system natural frequencies:

$$\omega_x = \omega_y = \sqrt{R^*} = \sqrt{1-\frac{\alpha}{\sqrt{x^{*2}+(1+z^*)^2}}}, \quad \omega_z = 1. \tag{15}$$

From the results of the equilibrium analysis of Eq. (7), we can estimate the system linear parameters based on the light sphere experiment ($M^* = 0.76$, $L^* = 8.93$) in Govardhan and Williamson (1997).

In order to estimate the structural pretension α from Eq. (10), we need experimental data for the mean sphere position x^* and mean drag β_D. However Govardhan and Williamson (1997) only documented information on β_D. We thus estimate the mean position x^* by employing the light spherical pendulum model studied in Shoshani and Gottlieb (2011):

$$x^* = \frac{\sqrt{2}}{2}\sqrt{1-\sqrt{1-\left(\frac{2\eta u^2\beta_D}{\mu}\right)}}, \tag{16}$$

where $\eta = 0.067$ from Eq. (8), $\mu = (1-M^*)(M^*+C_A) = 0.19$.

Note that Eq. (16) is only valid for $\mathrm{Re} < 12\,000$ due to the infinite stiffness of the rigid-body pendulum model. Therefore we linearly extrapolate x^* derived from Eq. (16) that satisfy $6000 < \mathrm{Re} < 8000$, and compute its modified values for $8000 < \mathrm{Re} < 13\,000$. The experimental results and the calculated equilibrium position x^* for Govardhan and Williamson (1997) are presented in Fig. 2.

Given β_D and x^*, we can estimate the tether stiffness α from the first equation in Eq. (10) for a given mean lift coefficient β_L. Following the analysis of Lee et al. (2013) on a neutrally buoyant tethered sphere, in which the lift at $\mathrm{Re}=300$ is very small (ranging from -0.0207 to 0.0028), we choose three small lift coefficients (-0.1, 0 and 0.1) to investigate the influence of β_L on α. The results in Fig. 3(b) show that the α values associated with $\beta_L = -0.1$ and $\beta_L = 0$ are reasonably close to 1, which is consistent with the description of the tethered sphere experiment (Govardhan and Williamson, 1997). Moreover, in Behara et al. (2011) it was demonstrated that the transverse force coefficient $C_{y,\mathrm{rms}}$ decreased strongly as a function of reduced velocity $U^* = U_\infty/f_n d$ from 0.12 to very close to zero (0.02) in the range of $5 < U^* < 9$. Therefore we assume $\beta_L = 0$ throughout the analysis, as the range of reduced velocity investigated here are between $4 < U^* < 40$.

The average value of the normalized tether stretch α is $\alpha_{\mathrm{avg}} = 0.96, 0.97$ for the unmodified and modified estimates and the mean streamwise x position reconstructed by α_{avg} with Eq. (10) has a maximum error of 2.5% when compared with the results in Fig. 2(b). Thus from here on, we will use the constant value α_{avg} for α throughout our analysis. Substitution of α into the equation that defines the equilibrium length yields a tether stiffness of $k = 5.64$, 7.15 (N/m) and enables computation of the frequency scale $\omega_s = 11.93$, 13.42 for the unmodified and the modified estimates respectively. Thus, with α known, we obtain the vertical equilibrium position z^* from Eq. (10), which is plotted in Fig 4.

With z^* also known, the natural frequencies ω_x, ω_y can now be computed. Note that the natural frequency without aerodynamics can readily be obtained by setting x^* and z^* in Eq. (15) to zero ($\omega_{x,y} = \sqrt{1-\alpha}$) and serves as a lower bound for

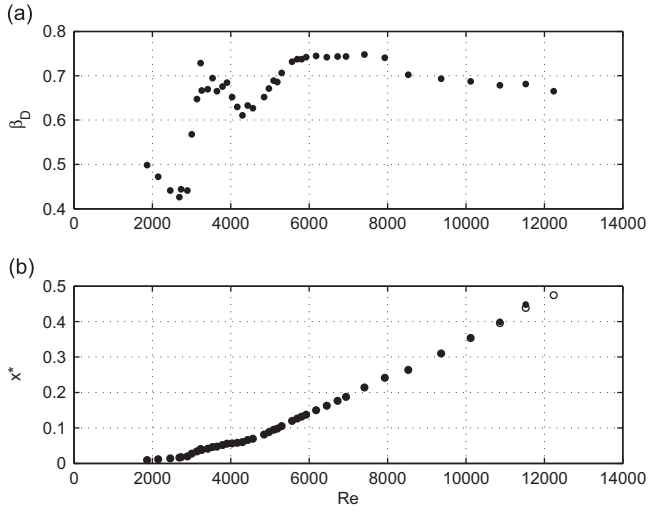


Fig. 2. (a) Measured mean drag β_D. (b) Mean sphere position in the streamwise direction x^*. ● computed from Eq. (16) $2000 < \mathrm{Re} < 12000$, ○ extrapolated x^* based on Eq. (16).

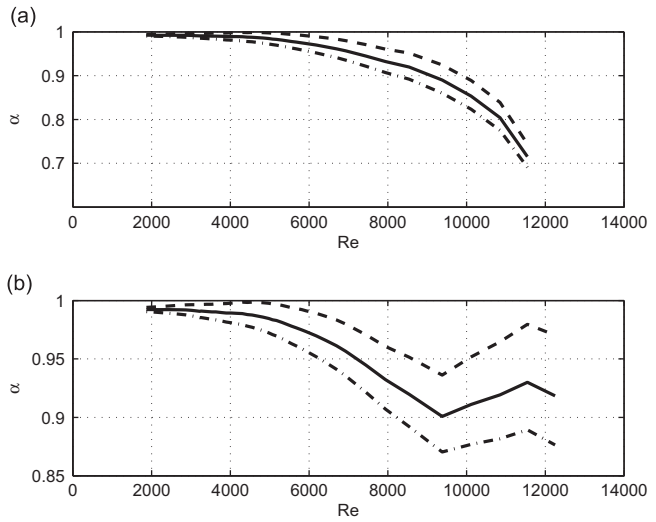


Fig. 3. Estimation of structural pretension α for several small values of lift coefficients. $--\beta_L=-0.1$, $--\beta_L=0$, -.- $\beta_L=0.1$. (a) Estimated from x^* from Eq. (16). (b) Estimated from extrapolated x^*.

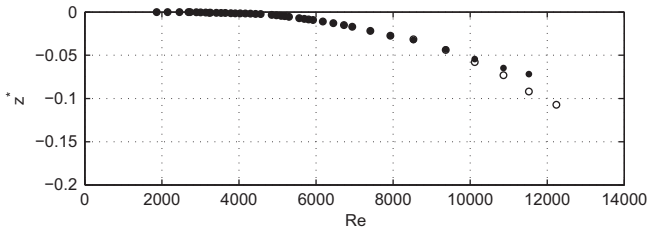


Fig. 4. The vertical equilibrium position z^* computed with experimental data and estimated linear parameters. Symbols ● and ○ are the same as in Fig. 2(b).

the frequencies with aerodynamics. The results shown in Fig. 5 further justify the assumption used to derive the reduced-order model ($1=\omega_z>\omega_x,\omega_y$).

We are interested in the dynamical system frequency ratios to determine possible conditions for internal resonances (Nayfeh and Mook, 1979). In Fig. 6(a) we show the frequency ratio between ω_z and ω_x as a function of reduced velocity, where we recall that $\omega_x=\omega_y$ in Eq. (15). Note that $\omega_z\approx 5\omega_y$ for $U^*=20$ and that the ratio decreases to $\omega_z\approx 3\omega_y$ for $U^*=40$. Also from the normalized frequency $f^*=f_{\mathrm{LC}}/f_n$ documented in Govardhan and Williamson (2005) and Govardhan and Williamson (1997) (where f_{LC} is the measured transverse limit-cycle frequency and f_n is the transverse natural frequency), we can infer the frequency ratio between ω_q and ω_y. This ratio is reproduced and extrapolated from $U^*=20$ to $U^*=40$ which corresponds to the maximal value documented in the tethered aerostat experiment (Coulomb-Pontbriand and Nahon, 2009). Note that $\omega_q\approx\omega_y$ in the range $4<U^*<8$ ($f^*\approx 1$), $\omega_q\approx 2\omega_y$ near $U^*=20$ ($f^*\approx 2$) and $\omega_q\approx 3\omega_y$($f^*\approx 3$) near $U^*=30$.

Thus, we conclude that internal resonance conditions for the tethered sphere in our range of interest ($4<U^*<40$) are $\omega_y:\omega_x=1:1$ and $\omega_q:\omega_y=m:1$ for $m=1,2,3$.

3.2. Stability of the reduced-order system

We rewrite Eq. (9) as a system of four first order equations:

$$\begin{aligned}
x_{1\tau}&=x_2\\
x_{2\tau}&=-\overline{R}x_1+\eta u^2x_3-\delta_1x_2\\
x_{3\tau}&=x_4\\
x_{4\tau}&=-\omega_q^2x_3+(\alpha_q-\gamma_qx_3^2)x_4+\mu_qx_2.
\end{aligned}\tag{17}$$

The natural frequencies and damping coefficients are identical to those of the full model. The equilibrium of Eq. (17) is $y^*=q^*=0$ and its stability is determined by the following characteristic polynomial:

$$\lambda^4+e_1\lambda^3+e_2\lambda^2+e_3\lambda+e_4=0,\tag{18}$$

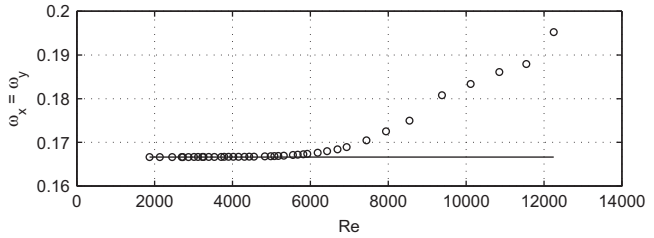


Fig. 5. The natural frequencies in the streamwise ω_x and transverse ω_y directions: – without aerodynamics, ○ from full model of the extrapolated version.

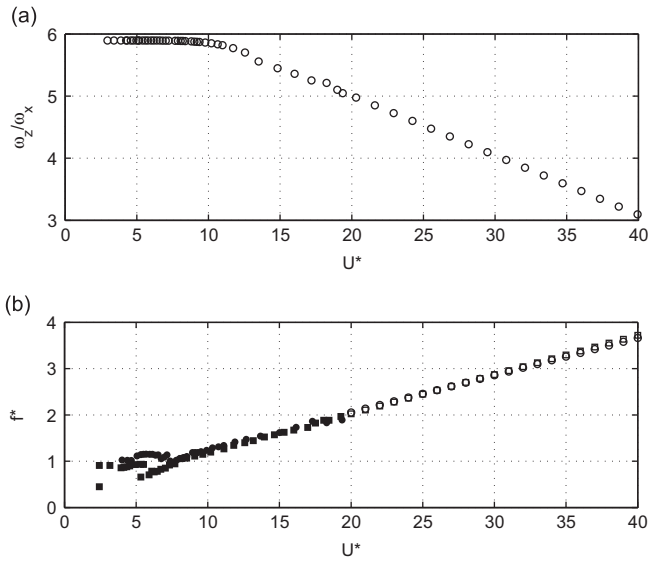


Fig. 6. (a) The frequency ratio between $\omega_z = 1$ and $\omega_x = \omega_y$ as a function of reduced velocity U^*. (b) The normalized transverse limit-cycle frequency f^* reproduced (solid symbols) and linearly extrapolated (hollow symbols) from Govardhan and Williamson (2005) for $M^* = 0.45, L^* = 13.8$ (□) and Govardhan and Williamson (1997) for $M^* = 0.76, L^* = 8.93$ (○).

where

$$e_1 = -\alpha_q + \delta_1, \quad e_2 = \omega_q^2 - \alpha_q\delta_1 + \omega_y^2, \quad e_3 = -\mu_q\eta u^2 + \delta_1\omega_q^2 - \alpha_q\omega_y^2, \quad e_4 = \omega_q^2\omega_y^2. \tag{19}$$

Following the Routh–Hurwitz criteria (Meirovitch, 1988), the condition for asymptotic stability of the equilibrium requires positive coefficients in Eq. (18) ($e_i > 0$ for $i = 1, 2, 3, 4$) and a positive third Hurwitz determinant:

$$\delta_3^e = e_1e_2e_3 - e_3^2 - e_4e_1^2 > 0. \tag{20}$$

Since the characteristic polynomial cannot have zero eigenvalues ($e_4 > 0$), the only stability threshold is that of flutter (or a Hopf bifurcation) where there is a single pair of pure imaginary eigenvalues ($\lambda_{1,2} = \pm i\Omega_H$) among the roots of Eq. (18). The Hopf threshold can be determined by setting δ_3^e in Eq. (20) to zero augmented by the requirement $e_3/e_1 > 0$ as the Hopf frequency is defined by $\Omega_H = \sqrt{e_3/e_1}$.

Based on the experimental data in Govardhan and Williamson (1997), we will use the Hopf threshold and the reduced-order model to deduce the linear wake oscillator parameters. We write the condition for the Hopf threshold ($\delta_3^e = 0$) as a simple quadratic polynomial equation for μ_q:

$$C_2\mu_q^2 + C_1\mu_q + C_0 = 0, \tag{21}$$

where

$$\begin{aligned} C_2 &= -(\eta u_H^2)^2 \\ C_1 &= \eta u_H^2\left(\alpha_q\delta_1(\delta_1 - \alpha_q) + (\alpha_q + \delta_1)(\omega_{q,H}^2 - \omega_{y,H}^2)\right) \\ C_0 &= -\alpha_q\delta_1\left((\omega_{y,H}^2 - \omega_{q,H}^2)^2 + (\alpha_q - \delta_1)(\alpha_q\omega_{y,H}^2 - \delta_1\omega_{q,H}^2)\right). \end{aligned} \tag{22}$$

The subscript H indicates the quantities at the Hopf threshold, which was measured in Govardhan and Williamson (1997) as Re=2450. Thus, u_H is computed given known values for η and ω_y. Secondly, following the measurements (Govardhan and

Williamson, 1997, 2005) of $f^* \approx 1$ in the range $4 < U^* < 13$, we assume that the wake natural frequency ω_q is equal to the transverse limit-cycle frequency $\omega_{y,\mathrm{LC}}$. Therefore the value of ω_q at the Hopf threshold can be deduced by utilizing the measured $\omega_{y,\mathrm{LC}}$ in Govardhan and Williamson (1997). Consequently, the only three unknowns in Eq. (21) are α_q, μ_q and δ.

We assume that the structural damping ratio is in the range $0.001 < \zeta < 0.1$. For a given δ where $\delta = 2\zeta\omega_y$, there is a direct relationship between α_q and μ_q. We choose an appropriate set of parameters from the relationship according to later criteria. Fig. 7 portrays the possible combinations of α_q and μ_q for different δ values.

In order to validate the selected linear parameter set, we substitute their values into Eq. (20) to calculate the Huruwitz third determinant associated with the transverse direction and the Hopf frequency ($\Omega_H = \sqrt{e_3/e_1}$). Indeed the result has a zero-crossing at the measured Hopf threshold Re=2450 where $\Omega_H = 0.033$.

It is instructive to compare the Hopf frequency Ω_H normalized by the transverse natural frequency ω_y to the measured normalized limit-cycle frequency f^* in Govardhan and Williamson (1997). Fig. 8 depicts both the measured f^* in Govardhan and Williamson (1997) and the estimated normalized Hopf frequency $\Omega_H^* = \Omega_H/\omega_y$. It is clear from Fig. 8 that Ω_H^* is a lower bound for the limit-cycle frequency beyond the Hopf threshold, and as such further validates the estimated linear parameter set.

The key points resulting from the equilibrium analysis include: (*i*) the mean drag β_D, lift β_L and tether pretension α coefficients can be deduced from Eq. (10) given independent measurements of the equilibrium positions x^*, z^* for increasing values of normalized flow velocity. Alternatively one can make use of the limiting case analysis given the mean drag and an estimate for the mean lift; and (*ii*) the linear structural damping δ, interaction coupling μ_q and linear wake α_q coefficients

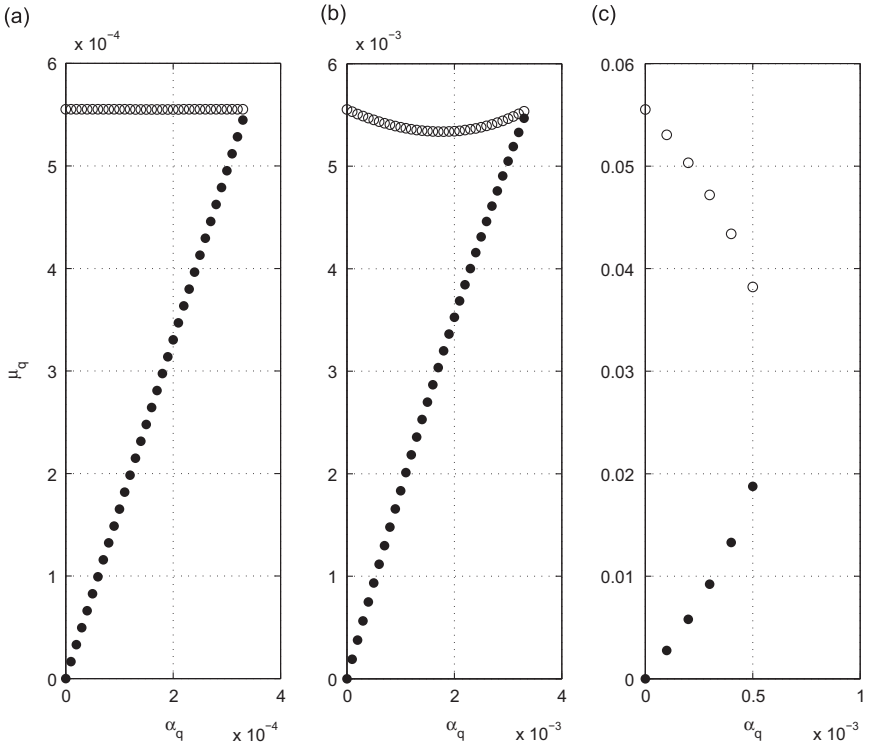


Fig. 7. Possible combinations of α_q and μ_q. (a) $\zeta = 0.001$, $\delta = 3.36e-4$. (b) $\zeta = 0.01$, $\delta = 3.36e-3$. (c) $\zeta = 0.1$, $\delta = 3.36e-2$.

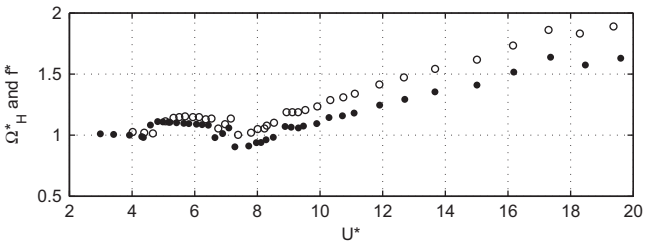


Fig. 8. The comparison between the normalized Hopf frequency Ω_H^* (●) and the measured normalized transverse limit-cycle frequency f^* (○) in Govardhan and Williamson (1997).

can readily be estimated by making use of the Routh–Huruwitz criteria at the Hopf threshold Eq. (21). We note that this parameter estimation methodology is not unique, but can be validated by comparison of the normalized Hopf frequency and the independently measured limit-cycle frequency near the VIV threshold.

4. Asymptotic and numerical analysis of a tethered sphere

The objective of the asymptotic analysis in this section is to extend the model-based estimation methodology to evaluate the nonlinear system parameters given sufficient data from the periodic response of the tethered sphere. The asymptotic multiple-scales method is applied first to the reduced order system Eq. (9) and then to the full system Eq. (7) where we consider two distinct cases for internal resonances based on the measured limit-cycle frequency. Specifically in subsection 4.1, the cubic wake parameter γ_q is obtained from the reduced-order system analysis; in subsection 4.2, the quadratic wake coupling coefficients are determined by the full system analysis. In Subsection 4.3, we construct the system bifurcation structure augmented by the numerical time-series, state space projections and Poincaré maps to demonstrate the agreement with the benchmark experiment.

4.1. Multiple scales of the reduced-order system

The main purpose of applying the asymptotic multiple scales method to the reduced-order model is to deduce the cubic wake parameter γ_q. We expand the nonlinear restoring force in Eq. (9) up to cubic order to yield a coupled Duffing–van der Pol dynamical system:

$$\begin{aligned} &y_{\tau\tau}+\omega_y^2 y=\eta u^2 q-\delta y_\tau-\tilde{\alpha}y^3\\ &q_{\tau\tau}+\omega_q^2 q=(\alpha_q-\gamma_q q^2)q_\tau+\mu_q y_\tau, \end{aligned} \tag{23}$$

where the Duffing constant is $\tilde{\alpha}=\alpha/2(x^{*2}+(1+z^*)^2)^{3/2}$. Due to the property of $\overline{R}$ in Eq. (9) the expansion has no even power contribution. Following Nayfeh and Mook (1979), we assume a small amplitude asymptotic solution form:

$$y=\epsilon y_1+\epsilon^2 y_2+\epsilon^3 y_3,\quad q=\epsilon q_1+\epsilon^2 q_2+\epsilon^3 q_3, \tag{24}$$

where ϵ is a small parameter. We define the time scales by $T_i=\epsilon^i\tau$ for $i=0,1,2$ where the transverse displacement y and the dynamic wake q are functions of fast T_0 and slow T_1, T_2 times (eg. $y_i=y_i(T_0,T_1,T_2)$, $q_i=q_i(T_0,T_1,T_2)$ for $i=1,2,3\ldots$).

We define the following coefficients via the small parameter:

$$\eta=\epsilon^2\hat{\eta},\quad ,\delta=\epsilon^2\hat{\delta}_1,\quad ,\alpha_q=\epsilon^2\hat{\alpha}_q,\quad ,\mu_q=\epsilon^2\hat{\mu}_q,\quad ,\gamma_q=\hat{\gamma}_q. \tag{25}$$

Substitution of the small amplitudes Eq. (24) and small parameters Eq. (25) into the truncated system in Eq. (23) yields the following set of recursive equations for three orders of ϵ:

$$\begin{aligned} O(\epsilon^1):\quad &D_0^2 y_1+\omega_y^2 y_1=0\\ &D_0^2 q_1+\omega_q^2 q_1=0, \end{aligned} \tag{26}$$

$$\begin{aligned} O(\epsilon^2):\quad &D_0^2 y_2+\omega_y^2 y_2=-2D_0D_1y_1\\ &D_0^2 q_2+\omega_q^2 q_2=-2D_0D_1q_1, \end{aligned} \tag{27}$$

$$\begin{aligned} O(\epsilon^3):\quad &D_0^2 y_3+\omega_y^2 y_3=-2D_0D_2y_1-D_1^2y_1-2D_0D_1y_2\\ &\qquad\qquad +\hat{\eta}u^2q_1-\hat{\delta}D_0y_1-\tilde{\alpha}y_1^3\\ &D_0^2 q_3+\omega_q^2 q_3=-2D_0D_2q_1-D_1^2q_1-2D_0D_1q_2\\ &\qquad\qquad +(\hat{\alpha}_q-\hat{\gamma}_q q_1^2)D_0q_1+\hat{\mu}_q D_0y_1. \end{aligned} \tag{28}$$

where D_i denotes $\partial/\partial T_i$ for i=0,1,2. The solution of Eq. (26) is

$$y_1=B_1(T_1,T_2)e^{i\omega_y T_0}+\text{cc},\quad q_1=Q_1(T_1,T_2)e^{i\omega_q T_0}+\text{cc}. \tag{29}$$

where the amplitudes B_1, Q_1 are functions of slow times and cc stands for complex conjugate. Omitting the complex conjugate notation for simplicity and substitution of Eq. (29) into Eq. (27) yields

$$D_0^2 y_2+\omega_y^2 y_2=-2i\omega_y D_1B_1e^{i\omega_y T_0},\quad D_0^2 q_2+\omega_q^2 q_2=-2i\omega_q D_1Q_1e^{i\omega_q T_0}. \tag{30}$$

In order to eliminate secular terms, we require that $D_1\,B_1=D_1\,Q_1=0$. Thus, B_1 and Q_1 do not depend on the slow time T_1 and the particular nonsecular solution of Eq. (30) is $y_2=q_2=0$. Substitution of Eq. (29) and ($y_2=q_2=0$) into Eq. (28) yields

$$\begin{aligned} D_0^2 y_3+\omega_y^2 y_3=&-2i\omega_y D_2B_1e^{i\omega_y T_0}+\hat{\eta}u^2Q_1e^{i\omega_q T_0}-i\omega_y\hat{\delta}_1B_1e^{i\omega_y T_0}\\ &-3\tilde{\alpha}|B_1|^2B_1e^{i\omega_y T_0}-\tilde{\alpha}B_1^3e^{3i\omega_y T_0} \end{aligned}$$

$$D_0^2 q_3 + \omega_q^2 q_3 = -2i\omega_q D_2 Q_1 e^{i\omega_q T_0} + i\omega_q(\hat{\alpha}_q - \hat{\gamma}_q|Q_1|^2)Q_1 e^{i\omega_q T_0} + i\omega_y \hat{\mu}_q B_1 e^{i\omega_y T_0} - i\omega_q \hat{\gamma}_q Q_1^3 e^{3i\omega_q T_0}. \tag{31}$$

For the case of a 1:1 internal resonance which is valid in the range $4 < U^* < 13$ (see Fig. 6(b) where $f^* \approx 1$), we assume a small detuning between the wake and the transverse natural frequencies:

$$\epsilon^2 \sigma_y = \omega_q - \omega_y. \tag{32}$$

Elimination of secular terms in Eq. (31) yields the following coupled slowly-varying evolution equations:

$$\begin{aligned} &-2i\omega_y D_2 B_1 + \hat{\eta} u^2 Q_1 e^{i\sigma_y T_2} - i\omega_y \hat{\delta}_1 B_1 - 3\tilde{\alpha}|B_1|^2 B_1 = 0 \\ &-2i\omega_q D_2 Q_1 + i\omega_q(\hat{\alpha}_q - \hat{\gamma}_q|Q_1|^2)Q_1 + i\omega_y \hat{\mu}_q B_1 e^{-i\sigma_y T_2} = 0. \end{aligned} \tag{33}$$

We make use of polar form for the slowly varying amplitudes $B_1 = a_y \exp(i\theta_y)/2$ and $Q_1 = a_q \exp(i\theta_q)/2$ to yield

$$\begin{aligned} D_2 a_y &= -\frac{1}{2}\hat{\delta}_1 a_y + \frac{1}{2\omega_y}\hat{\eta} u^2 a_q \sin\phi_y \\ D_2 \theta_y &= \frac{3}{8\omega_y}\tilde{\alpha} a_y^2 - \frac{1}{2\omega_y}\hat{\eta} u^2 \frac{a_q}{a_y}\cos\phi_y \\ D_2 a_q &= \frac{1}{2}\hat{\alpha}_q a_q - \frac{1}{8}\hat{\gamma}_q a_q^3 + \frac{1}{2}\hat{\mu}_q \frac{\omega_y}{\omega_q} a_y \cos\phi_y \\ D_2 \theta_q &= -\frac{1}{2}\hat{\mu}_q \frac{\omega_y a_y}{\omega_q a_q}\sin\phi_y. \end{aligned} \tag{34}$$

where $\phi_y = \theta_q - \theta_y + \sigma_y T_2$. Eq. (34) is then rewritten in its final form which consists of the slow variation of the transverse displacement a_y, the wake amplitude a_q and their phase difference ϕ_y:

$$\begin{aligned} D_2 a_y &= -\frac{1}{2}\hat{\delta}_1 a_y + \frac{1}{2\omega_y}\hat{\eta} u^2 a_q \sin\phi_y \\ D_2 a_q &= \frac{1}{2}\hat{\alpha}_q a_q - \frac{1}{8}\hat{\gamma}_q a_q^3 + \frac{1}{2}\hat{\mu}_q \frac{\omega_y}{\omega_q} a_y \cos\phi_y \\ D_2 \phi_y &= -\frac{1}{2}\hat{\mu}_q \frac{\omega_y a_y}{\omega_q a_q}\sin\phi_y + \frac{1}{2\omega_y}\hat{\eta} u^2 \frac{a_q}{a_y}\cos\phi_y - \frac{3}{8\omega_y}\tilde{\alpha} a_y^2 + \sigma_y. \end{aligned} \tag{35}$$

Setting the slow time derivatives in Eq. (35) to zero enables elimination of the phase variable ϕ_y, which results in two algebraic equations for the two unknown amplitudes a_y and a_q and one unknown parameter γ_q. We make use of the measured transverse limit-cycle amplitude in Govardhan and Williamson (1997) to determine a_y. Hence we are able to express the two algebraic equations in terms of the two unknowns a_q and γ_q (see Appendix A), which can be further manipulated to yield a single bi-quartic equation for a_q:

$$e_4 a_q^8 + e_3 a_q^6 + e_2 a_q^4 + e_1 a_q^2 + e_0 = 0, \tag{36}$$

where

$$\begin{aligned} &e_4 = -c_3 d_1^2 + c_2 d_1 d_2 - c_1 d_2^2, \quad e_3 = -c_4 d_1^2 + c_2 d_1 d_3 - 2c_1 d_2 d_3, \\ &e_2 = -c_5 d_1^2 - c_1 d_3^2 + c_2 d_1 d_4 - 2c_1 d_2 d_4, \quad e_1 = -2c_1 d_3 d_4, \quad e_0 = -c_1 d_4^2. \end{aligned} \tag{37}$$

and c_i and d_i are in Appendix A. Eq. (36) has four roots for a_q^2, which are obtained numerically. Two roots are real and the small real solution is selected for a_q. Substitution of the selected solution for a_q^2 into (56a,b-b) yields the estimate for the cubic van der Pol coefficient γ_q assuming a zero detuning $\sigma_y = 0$. By selecting an appropriate structural damping ratio $\zeta = 0.08$, we get a linear damping coefficient $\delta = 2.69e-2$ and are able to reconstruct the measured transverse limit-cycle amplitude in Govardhan and Williamson (1997) with the full model in Eq. (7) where the wake coupling coefficients k_i are set to zero. Fig. 9 portrays the estimated van der Pol coefficient γ_q and the numerically obtained transverse limit-cycle amplitude $|y|_{LC}$ using the estimated linear and nonlinear parameters.

In Fig. 9(a) the first few points deviate from the smooth curve due to the relatively large measurement error when the limit-cycle amplitude is small. Also observe in Fig. 9(b) that beyond Re=8000, the numerical result diverges from the experimental data. This is because the detuning in Eq. (32) no longer holds, as can be seen from the measured transverse limit-cycle frequencies in Govardhan and Williamson (1997).

4.2. Multiple scales of the full system

We translate the dynamical system in Eq. (7) about its equilibrium (x^*, z^*) and define new variables $s = x - x^*$, $w = z - z^*$. We then expand the nonlinear conservative restoring force terms to cubic order (see Appendix B):

$$\begin{aligned} &s_{\tau\tau} + k_{s1} s + k_{w1} w = \eta u^2 p - \delta s_\tau - (k_{ss1} s^2 + k_{yy1} y^2 + k_{ww1} w^2 + k_{sw1} sw) \\ &y_{\tau\tau} + \omega_y^2 y = \eta u^2 q - \delta y_\tau - (k_{sy2} sy + k_{yw2} yw) \end{aligned}$$

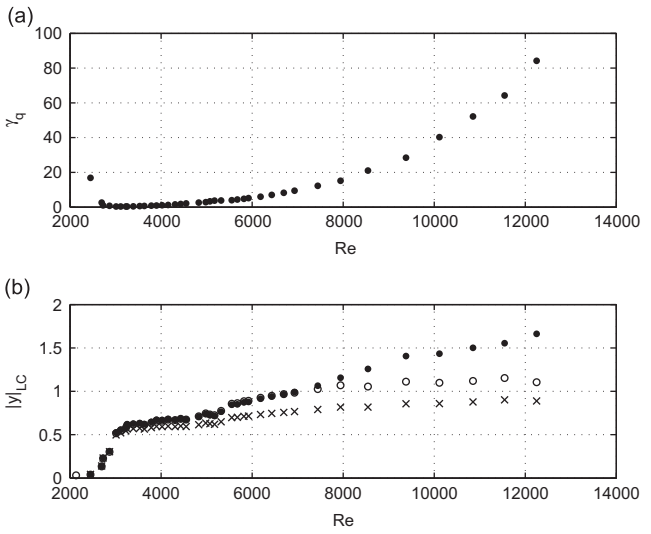


Fig. 9. (a) The estimated γ_q with $\delta = 2.69e-2$, $\alpha_{p,q,r} = 7e-4$, $\mu_{p,q,r} = 2.62e-2$. (b) The transverse limit-cycle amplitude. ○ measured in Govardhan and Williamson (1997) ● computed from the full model with zero wake coupling coefficients, × computed from the reduced-model.

$$
\begin{aligned}
&w_{\tau\tau} + k_{w3}w + k_{s3}s = \eta u^2 r - \delta w_\tau - \left(k_{ss3}s^2 + k_{yy3}y^2 + k_{ww3}w^2 + k_{sw3}sw\right)\\
&p_{\tau\tau} + \omega_p^2 p = (\alpha_p - \gamma_p p^2)p_\tau - (k_{p2}qq_\tau + k_{p3}q_\tau^2) + \mu_p x_\tau\\
&q_{\tau\tau} + \omega_q^2 q = (\alpha_q - \gamma_q q^2)q_\tau - (k_{q2}pq_\tau + k_{q3}p_\tau q + k_{q4}p_\tau q_\tau)\\
&\qquad\qquad - (k_{q6}rq_\tau + k_{q7}r_\tau q + k_{q8}r_\tau q_\tau) + \mu_q y_\tau\\
&r_{\tau\tau} + \omega_r^2 r = (\alpha_r - \gamma_r r^2)r_\tau - (k_{r2}qq_\tau + k_{r3}q_\tau^2) + \mu_r z_\tau,
\end{aligned}
\tag{38}
$$

where the coefficients k_{ijk} are defined in Appendix B and are shown to be of similar order in the range $2000 < \mathrm{Re} < 8000$.

We assume that small amplitude solution forms for the displacement s, y, w and the wake p, q, r where the variables are functions of fast (T_0) and slow ($T_1 = \epsilon\tau$) times:

$$
\begin{aligned}
&s = \epsilon s_1 + \epsilon^2 s_2, \quad y = \epsilon y_1 + \epsilon^2 y_2, \quad w = \epsilon w_1 + \epsilon^2 w_2\\
&p = \epsilon p_1 + \epsilon^2 p_2, \quad q = \epsilon q_1 + \epsilon^2 q_2, \quad r = \epsilon r_1 + \epsilon^2 r_2.
\end{aligned}
\tag{39}
$$

We define the following small parameters:

$$
\begin{aligned}
&\eta = \epsilon\hat{\eta}, \quad \delta = \epsilon\hat{\delta}, \quad k_{w1} = \epsilon\hat{k}_{w1}, \quad k_{s3} = \epsilon\hat{k}_{s3}\\
&\alpha_i = \epsilon\hat{\alpha}_i, \quad \mu_i = \hat{\mu}_i, \quad \gamma_i = \hat{\gamma}_i/\epsilon \quad \text{for} i = p, q, r.
\end{aligned}
\tag{40}
$$

Substitution of Eq. (39) and Eq. (40) into Eq. (38) and equating coefficients of like powers of ϵyields the following set of recursive equations for two orders of ϵ:

$$
\begin{aligned}
O(\epsilon^1):\quad & D_0^2 s_1 + \omega_s^2 s_1 = 0, \quad D_0^2 y_1 + \omega_y^2 y_1 = 0, \quad D_0^2 w_1 + \omega_w^2 w_1 = 0\\
& D_0^2 p_1 + \omega_p^2 p_1 = 0, \quad D_0^2 q_1 + \omega_q^2 q_1 = 0, \quad D_0^2 r_1 + \omega_r^2 r_1 = 0,
\end{aligned}
\tag{41}
$$

$$
\begin{aligned}
O(\epsilon^2):\quad & D_0^2 s_2 + \omega_s^2 s_2 = -2D_0D_1 s_1 + \hat{\eta}u^2 p_1 - \hat{\delta}D_0 s_1\\
&\qquad - (\hat{k}_{w1}w_1 + k_{ss1}s_1^2 + k_{yy1}y_1^2 + k_{ww1}w_1^2 + k_{sw1}s_1 w_1)\\
& D_0^2 y_2 + \omega_y^2 y_2 = -2D_0D_1 y_1 + \hat{\eta}u^2 q_1 - \hat{\delta}D_0 y_1 - (k_{sy2}s_1 y_1 + k_{yw2}y_1 w_1)\\
& D_0^2 w_2 + \omega_w^2 w_2 = -2D_0D_1 w_1 + \hat{\eta}u^2 r_1 - \hat{\delta}D_0 w_1\\
&\qquad - (\hat{k}_{s3}s_1 + k_{ss3}s_1^2 + k_{yy3}y_1^2 + k_{ww3}w_1^2 + k_{sw3}s_1 w_1)\\
& D_0^2 p_2 + \omega_p^2 p_2 = -2D_0D_1 p_1 + (\hat{\alpha}_p - \hat{\gamma}_p p_1^2)D_0 p_1 + \hat{\mu}_p D_0 s_1\\
&\qquad - (k_{p2}q_1 D_0 q_1 + k_{p3}(D_0 q_1)^2)\\
& D_0^2 q_2 + \omega_q^2 q_2 = -2D_0D_1 q_1 + (\hat{\alpha}_q - \hat{\gamma}_q q_1^2)D_0 q_1 + \hat{\mu}_q D_0 y_1\\
&\qquad - (k_{q2}p_1 D_0 q_1 + k_{q3}q_1 D_0 p_1 + k_{q4}D_0 p_1 D_0 q_1)\\
&\qquad - (k_{q6}r_1 D_0 q_1 + k_{q7}q_1 D_0 r_1 + k_{q8}D_0 q_1 D_0 r_1)\\
& D_0^2 r_2 + \omega_r^2 r_2 = -2D_0D_1 r_1 + (\hat{\alpha}_r - \hat{\gamma}_r r_1^2)D_0 r_1 + \hat{\mu}_r D_0 w_1\\
&\qquad - (k_{r2}q_1 D_0 q_1 + k_{r3}(D_0 q_1)^2).
\end{aligned}
\tag{42}
$$

Solution of the homogeneous equation (41) is

$$s_1 = A(T_1)e^{i\omega_s T_0} + \text{cc}, \quad y_1 = B(T_1)e^{i\omega_y T_0} + \text{cc}, \quad w_1 = C(T_1)e^{i\omega_w T_0} + \text{cc},$$
$$p_1 = P(T_1)e^{i\omega_p T_0} + \text{cc}, \quad q_1 = Q(T_1)e^{i\omega_q T_0} + \text{cc}, \quad r_1 = R(T_1)e^{i\omega_r T_0} + \text{cc}. \tag{43}$$

where the complex amplitudes are functions of slow time T_1. Substitution of Eq. (43) into Eq. (42) yields the following non-homogeneous equations:

$$\begin{aligned}
D_0^2 s_2 + \omega_s^2 s_2 = &-i\omega_s(\hat{\delta}_1 A + 2D_1 A)e^{i\omega_s T_0} + \hat{\eta} u^2 P e^{i\omega_p T_0} \\
&- \hat{k}_{w1} C e^{i\omega_w T_0} - k_{ss1} A^2 e^{2i\omega_s T_0} - k_{yy1} B^2 e^{2i\omega_y T_0} - k_{ww1} C^2 e^{2i\omega_w T_0} \\
&- k_{sw1} A C e^{i(\omega_s+\omega_w)T_0} - k_{sw1} A\overline{C} e^{i(\omega_s-\omega_w)T_0} \\
&- (k_{ss1}|A|^2 + k_{yy1}|B|^2 + k_{ww1}|C|^2) + \text{cc} \\
D_0^2 y_2 + \omega_y^2 y_2 = &-i\omega_y(\hat{\delta}_1 B + 2D_1 B)e^{i\omega_y T_0} + \hat{\eta} u^2 Q e^{i\omega_q T_0} \\
&- k_{sy2} A B e^{i(\omega_s+\omega_y)T_0} - k_{sy2} \overline{A} B e^{-i(\omega_s-\omega_y)T_0} \\
&- k_{yw2} B C e^{i(\omega_y+\omega_w)T_0} - k_{yw2} B\overline{C} e^{-i(\omega_w-\omega_y)T_0} + \text{cc} \\
D_0^2 w_2 + \omega_w^2 w_2 = &-i\omega_w(\hat{\delta}_1 C + 2D_1 C)e^{i\omega_w T_0} + \hat{\eta} u^2 R e^{i\omega_r T_0} \\
&- \hat{k}_{s3} A e^{i\omega_s T_0} - k_{ss3} A^2 e^{2i\omega_s T_0} - k_{yy3} B^2 e^{2i\omega_y T_0} - k_{ww3} C^2 e^{2i\omega_w T_0} \\
&- k_{sw3} A C e^{i(\omega_w+\omega_s)T_0} - k_{sw3} \overline{A} C e^{-i(\omega_s-\omega_w)T_0} \\
&- (k_{ss3}|A|^2 + k_{yy3}|B|^2 + k_{ww3}|C|^2) + \text{cc} \\
D_0^2 p_2 + \omega_p^2 p_2 = &\, i\omega_p(\hat{\alpha}_p P - \hat{\gamma}_p|P|^2 P - 2D_1 P)e^{i\omega_p T_0} + i\omega_s \hat{\mu}_p A e^{i\omega_s T_0} \\
&- i\omega_p \hat{\gamma}_p P^3 e^{3i\omega_p T_0} + (-i\omega_q k_{p2} + \omega_q^2 k_{p3})Q^2 e^{2i\omega_q T_0} - \omega_q^2 k_{p3}|Q|^2 + \text{cc} \\
D_0^2 q_2 + \omega_q^2 q_2 = &\, i\omega_q(\hat{\alpha}_q Q - \hat{\gamma}_q|Q|^2 Q - 2D_1 Q)e^{i\omega_q T_0} + i\omega_y \hat{\mu}_q B e^{i\omega_y T_0} \\
&- i\omega_q \hat{\gamma}_q Q^3 e^{3i\omega_q T_0} + (-i\omega_p k_{q3} - i\omega_q k_{q2} + \omega_p\omega_q k_{q4})PQ e^{i(\omega_p+\omega_q)T_0} \\
&+ (-i\omega_p k_{q3} + i\omega_q k_{q2} - \omega_p\omega_q k_{q4})P\overline{Q} e^{i(\omega_p-\omega_q)T_0} \\
&+ (-i\omega_q k_{q6} - i\omega_r k_{q7} + \omega_q\omega_r k_{q8})QR e^{i(\omega_q+\omega_r)T_0} \\
&+ (-i\omega_q k_{q6} + i\omega_r k_{q7} - \omega_q\omega_r k_{q8})Q\overline{R} e^{i(\omega_q-\omega_r)} + \text{cc} \\
D_0^2 r_2 + \omega_r^2 r_2 = &\, i\omega_r(\hat{\alpha}_r R - \hat{\gamma}_r|R|^2 R - 2D_1 R)e^{i\omega_r T_0} + i\omega_w \hat{\mu}_r C e^{i\omega_w T_0} \\
&- i\omega_r \hat{\gamma}_r R^3 e^{3i\omega_r T_0} + (-i\omega_q k_{r2} + \omega_q^2 k_{r3})Q^2 e^{2i\omega_q T_0} - \omega_q^2 k_{r3}|Q|^2 + \text{cc}.
\end{aligned} \tag{44}$$

For the case of nonresonant interactions, the solvability conditions of Eq. (44) reduce to the following decoupled set of evolution equations:

$$\begin{aligned}
&D_1 A + \frac{1}{2}\hat{\delta}A = 0, \quad D_1 P - \frac{1}{2}\hat{\alpha}_p P + \frac{1}{2}\hat{\gamma}_p|P|^2 P = 0 \\
&D_1 B + \frac{1}{2}\hat{\delta}B = 0, \quad D_1 Q - \frac{1}{2}\hat{\alpha}_q Q + \frac{1}{2}\hat{\gamma}_q|Q|^2 Q = 0 \\
&D_1 C + \frac{1}{2}\hat{\delta}C = 0, \quad D_1 R - \frac{1}{2}\hat{\alpha}_r R + \frac{1}{2}\hat{\gamma}_r|R|^2 R = 0,
\end{aligned} \tag{45}$$

which are valid below the Hopf threshold (see Section 3).

4.2.1. *Internal resonance of 1:1:6:2:1:1*

Recall that the equilibrium analysis in Subsection 3.1 revealed that the ratio between the wake and transverse natural frequencies $\omega_q{:}\,\omega_y = 1{:}\,1$ ($3 < U^* < 13$). Furthermore, recall that $\omega_x = \omega_y$, $\omega_z \approx 6\omega_x$ (see Fig. 6(a)) and that $\omega_p = 2\omega_q$. Thus for the case of an internal resonance between the structure and wake $\omega_s{:}\,\omega_y{:}\,\omega_w{:}\,\omega_p{:}\,\omega_q{:}\,\omega_r = 1{:}\,1{:}\,6{:}\,2{:}\,1{:}\,1$, we identify two detuning coefficients demonstrating the nearness of the transverse wake to its displacement (1:1) and the period doubled relationship between the streamwise ω_p and the transverse ω_q wake components (2:1) that have been demonstrated experimentally for moving light tethered spheres (Williamson and Govardhan, 1997):

$$\epsilon\sigma_y = \omega_q - \omega_y, \quad \epsilon\sigma_p = 2\omega_q - \omega_p. \tag{46}$$

Substitution of Eq. (46) into Eq. (44) enables identification of secular terms:

$$\begin{aligned}
&D_1 A + \frac{1}{2}\hat{\delta}A = 0, \quad D_1 C + \frac{1}{2}\hat{\delta}C = 0, \quad D_1 R - \frac{1}{2}\hat{\alpha}_r R + \frac{1}{2}\hat{\gamma}_r|R|^2 R = 0, \\
&i\omega_y(\hat{\delta}B + 2D_1 B) - \hat{\eta} u^2 Q e^{i\sigma_y T_1} = 0 \\
&i\omega_p(\hat{\alpha}_p P - \hat{\gamma}_p|P|^2 P - 2D_1 P) - (i\omega_q k_{p2} - \omega_q^2 k_{p3})Q^2 e^{i\sigma_p T_1} = 0 \\
&i\omega_q(\hat{\alpha}_q Q - \hat{\gamma}_q|Q|^2 Q - 2D_1 Q) + i\omega_y \hat{\mu}_q B e^{-i\sigma_y T_1} \\
&\quad - (i\omega_p k_{q3} - i\omega_q k_{q2} + \omega_p\omega_q k_{q4})P\overline{Q} e^{-i\sigma_p T_1} = 0,
\end{aligned} \tag{47a–f}$$

In Eq. (47a–f), we note that the streamwise A and vertical C displacements, and the wake in vertical direction R are uncoupled from the other variables. Therefore, we are only interested in the remaining equations for the transverse displacement B and wake amplitudes P, Q. We make use of the polar form $A_j=\frac{1}{2}a_j\exp(i\theta_j)$ for $j=y,p,q$ and substitute the solution into Eq. ((47a–f)b,d,e). Separating real from imaginary parts yields six slowly varying evolution equations for the amplitudes a_y,a_p,a_q and angles $\theta_y,\theta_p,\theta_q$ (see Appendix C). We then make use of the phase differences between the angles to rewrite the slowly varying system in its final form of limit-cycle amplitudes and phase differences:

$$
\begin{aligned}
D_1a_y &= -\frac{1}{2}\hat{\delta}_1a_y+\frac{1}{2}\frac{\hat{\eta}}{\omega_y}u^2a_q\sin\phi_y\\
D_1a_p &= \frac{1}{2}\hat{\alpha}_pa_p-\frac{1}{8}\hat{\gamma}_pa_p^3-\frac{1}{4}\frac{\omega_q}{\omega_p}k_{p2}a_q^2\cos\phi_p+\frac{1}{4}\frac{\omega_q^2}{\omega_p}k_{p3}a_q^2\sin\phi_p\\
D_1a_q &= \frac{1}{2}\hat{\alpha}_qa_q-\frac{1}{8}\hat{\gamma}_qa_q^3+\frac{1}{2}\frac{\omega_y}{\omega_q}\hat{\mu}_qa_y\cos\phi_y\\
&\quad -\frac{1}{4}\left(\frac{\omega_p}{\omega_q}k_{q3}-k_{q2}\right)a_pa_q\cos\phi_p+\frac{1}{4}\omega_pk_{q4}a_pa_q\sin\phi_p\\
D_1\phi_y &= \frac{1}{4}\left(\frac{\omega_p}{\omega_q}k_{q3}-k_{q2}\right)a_p\sin\phi_p+\frac{1}{4}\omega_pk_{q4}a_p\cos\phi_p\\
&\quad -\frac{1}{2}\frac{\omega_y}{\omega_q}\hat{\mu}_q\frac{a_y}{a_q}\sin\phi_y+\frac{1}{2}\frac{\hat{\eta}}{\omega_y}u^2\frac{a_q}{a_y}\cos\phi_y+\sigma_y\\
D_1\phi_p &= \frac{1}{2}\left(\frac{\omega_p}{\omega_q}k_{q3}-k_{q2}\right)a_p\sin\phi_p+\frac{1}{2}\omega_pk_{q4}a_p\cos\phi_p-\frac{\omega_y}{\omega_q}\hat{\mu}_q\frac{a_y}{a_q}\sin\phi_y\\
&\quad +\frac{\omega_qa_q^2}{4\omega_pa_p}\left(k_{p2}\sin\phi_p+\omega_qk_{p3}\cos\phi_p\right)+\sigma_p,
\end{aligned}
\tag{48}
$$

where $\phi_y=\theta_q-\theta_y+\sigma_yT_1$, $\phi_p=2\theta_q-\theta_p+\sigma_pT_1$. Setting the slowly varying derivatives to zero, we obtain five algebraic equations with five unknown state variables $a_y,a_p,a_q,\phi_y,\phi_p$ and seven unknown parameters γ_p,γ_q, k_p's and k_q's (see Appendix C).

After elimination of the phases, we are left with three algebraic equations with three unknown state variables a_y, a_p, a_q and seven unknown parameters. As before we make use of the measured transverse limit-cycle amplitude in Govardhan and Williamson (1997) to determine a_y. In addition, we make use of computed a_q and γ_q from the reduced-order asymptotic analysis in Section 4.1. The streamwise wake amplitude a_p is obtained from the numerical integration of the full model without the wake coupling terms. Therefore the unknowns are only the wake coupling parameters k_p's and k_q's in Eq. (7). Following Shoshani and Gottlieb (2011), we set the two less significant parameters k_{p2} and k_{p3} to 1 and express the three algebraic equations in terms of the k_q's as follows:

$$
\begin{aligned}
&m_{11}k_{q2}^2+m_{12}k_{q3}^2+m_{13}k_{q4}^2+m_{14}k_{q2}k_{q3}+m_{15}k_{q2}k_{q4}+m_{16}k_{q3}k_{q4}\\
&\quad +m_{17}k_{q2}+m_{18}k_{q3}+m_{19}k_{q4}+m_{10}=0\\
&m_{21}k_{q2}^2+m_{22}k_{q3}^2+m_{23}k_{q4}^2+m_{24}k_{q2}k_{q3}+m_{27}k_{q2}+m_{28}k_{q3}+m_{29}k_{q4}=0\\
&m_{31}k_{q2}^2+m_{32}k_{q3}^2+m_{33}k_{q4}^2+m_{34}k_{q2}k_{q3}+m_{37}k_{q2}+m_{38}k_{q3}+m_{39}k_{q4}+m_{30}=0,
\end{aligned}
\tag{49}
$$

where the coefficients are listed in Appendix D. Substituting $k_{q2}=-11,k_{q3}=6,k_{q4}=-14.5$ and the values of all the state variables and parameters at Re $=6000$, the residuals of the components in Eq. (49) are -1.31×10^{-11}, -1.56×10^{-7}, -4.90×10^{-11} respectively. We conduct a numerical continuation analysis where we find the continuously varying wake coupling parameters as a function of Re that best satisfy the experimental data. The results are shown in Fig. 10. Note that k_r's are set to be the same as the corresponding k_p's, whereas $k_{q6,7,8}$ are the opposite sign of $k_{q2,3,4}$. In the region $3000<\mathrm{Re}<8000$, the maximal error between the numerical solution and the measured data is 6.79%.

The comparison between the frequency ratio of the numerical simulation and the measured data is shown in Fig. 11. We note that the wake limit-cycle dynamics in the transverse direction a_q is close to half the frequency of the streamwise direction a_p and the frequency of the transverse sphere projection a_y is close to that of the transverse wake a_q (see Eq. (46)). Consequently the numerical analysis of f^* depicted in Fig. 11 shows that range of validity reduces to $4.8<U^*<9.6$.

4.2.2. *Internal resonance of 1:1:5:4:2:ω_r*

Based on the extrapolated f^* in Fig. 6(b), the ratio between the transverse limit-cycle frequency $\omega_{y,\mathrm{LC}}$ and the transverse natural frequency ω_y is about 2 for $U^*=20$. Thus we identify a second internal resonance condition where $\omega_s:\omega_y:\omega_w:\omega_p:\omega_q:\omega_r=1:1:5:4:2:\omega_r$. Therefore, we assume the following detunings: $\epsilon\sigma_1=\omega_q-\omega_y-\omega_s$ and $\epsilon\sigma_2=\omega_w-\omega_p-\omega_s$. Furthermore, as the Reynolds number increases, the magnitude of the parameter ηu^2 also increases. We define $\eta=\hat{\eta}$ in this case and keep the other small parameter assumptions in Eq. (40) and the small amplitude assumption in Eq. (39) unchanged. The order equations deduced from Eq. (38) are

$$
O(\epsilon^1):\quad D_0^2s_1+\omega_s^2s_1-\hat{\eta}u^2p_1=0,\quad D_0^2y_1+\omega_y^2y_1-\hat{\eta}u^2q_1=0,
$$

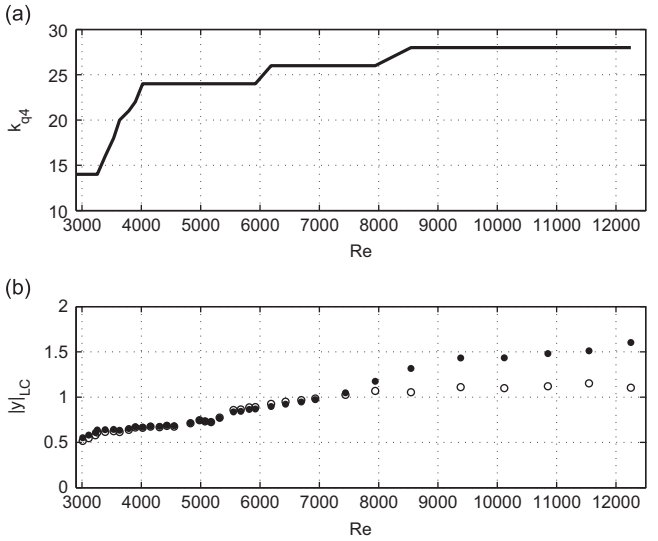


Fig. 10. (a) The wake coupling coefficients k_{q4}. (b) The numerical integration of the full system incorporating all the estimated parameters ● in comparison with the experimental data ○ from Govardhan and Williamson (1997).

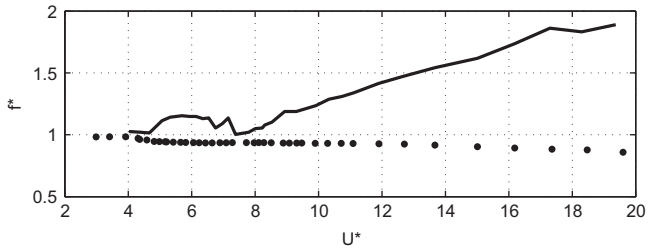


Fig. 11. The measured normalized frequency f^* in Govardhan and Williamson (1997) ● the numerically obtained normalized transverse limit-cycle frequency $\omega_{y,LC}/\omega_y$ where ω_y is the natural frequency computed according to (15).

$$
\begin{aligned}
&D_0^2 w_1+\omega_w^2 w_1-\hat{\eta}u^2 r_1=0\\
&D_0^2 p_1+\omega_p^2 p_1=0,\quad D_0^2 q_1+\omega_q^2 q_1=0,\quad D_0^2 r_1+\omega_r^2 r_1=0.
\end{aligned}
\tag{50}
$$

$$
\begin{aligned}
O(\epsilon^2):\quad D_0^2 s_2+\omega_s^2 s_2-\hat{\eta}u^2 p_2 &= -2D_0D_1s_1-\hat{\delta}_1 D_0 s_1\\
&\quad -(\hat{k}_{w1}w_1+k_{ss1}s_1^2+k_{yy1}y_1^2+k_{ww1}w_1^2+k_{sw1}s_1w_1)\\
D_0^2 y_2+\omega_y^2 y_2-\hat{\eta}u^2 q_2 &= -2D_0D_1y_1-\hat{\delta}_1 D_0 y_1\\
&\quad -(k_{sy2}s_1y_1+k_{yw2}y_1w_1)\\
D_0^2 w_2+\omega_w^2 w_2-\hat{\eta}u^2 r_2 &= -2D_0D_1w_1-\hat{\delta}_1 D_0 w_1\\
&\quad -(\hat{k}_{s3}s_1+k_{ss3}s_1^2+k_{yy3}y_1^2+k_{ww3}w_1^2+k_{sw3}s_1w_1)\\
D_0^2 p_2+\omega_p^2 p_2 &= -2D_0D_1p_1+(\hat{\alpha}_p-\hat{\gamma}_p p_1^2)D_0p_1+\hat{\mu}_p D_0 s_1\\
&\quad -(k_{p2}q_1D_0q_1+k_{p3}(D_0q_1)^2)\\
D_0^2 q_2+\omega_q^2 q_2 &= -2D_0D_1q_1+(\hat{\alpha}_q-\hat{\gamma}_q q_1^2)D_0q_1+\hat{\mu}_q D_0 y_1\\
&\quad -(k_{q2}p_1D_0q_1+k_{q3}q_1D_0p_1+k_{q4}D_0p_1D_0q_1)\\
&\quad -(k_{q6}r_1D_0q_1+k_{q7}q_1D_0r_1+k_{q8}D_0q_1D_0r_1)\\
D_0^2 r_2+\omega_r^2 r_2 &= -2D_0D_1r_1+(\hat{\alpha}_r-\hat{\gamma}_r r_1^2)D_0r_1+\hat{\mu}_r D_0 w_1\\
&\quad -(k_{r2}q_1D_0q_1+k_{r3}(D_0q_1)^2).
\end{aligned}
\tag{51}
$$

We note that the first order set of equations in Eq. (50) are strongly coupled. This hard excitation is due to the assumption of the increased magnitude of ηu^2. Thus, the homogeneous solution is

$$
\begin{aligned}
&s_1=A(T_1)e^{i\omega_s T_0}+H_sP(T_1)e^{i\omega_p T_0}+\text{cc},\quad p_1=P(T_1)e^{i\omega_p T_0}+\text{cc},\\
&y_1=B(T_1)e^{i\omega_y T_0}+H_yQ(T_1)e^{i\omega_q T_0}+\text{cc},\quad q_1=Q(T_1)e^{i\omega_q T_0}+\text{cc},
\end{aligned}
$$

$$w_1 = C(T_1)e^{i\omega_w T_0} + H_w R(T_1)e^{i\omega_r T_0} + \text{cc}, \quad r_1 = R(T_1)e^{i\omega_r T_0} + \text{cc}. \tag{52}$$

where

$$H_s = \frac{\hat{\eta}u^2}{\omega_s^2 - \omega_p^2}, \quad H_y = \frac{\hat{\eta}u^2}{\omega_y^2 - \omega_q^2}, \quad H_w = \frac{\hat{\eta}u^2}{\omega_w^2 - \omega_r^2}. \tag{53}$$

Substitution of (52) into (51) yields a standard set of non-homogeneous equations (see Appendix E). Identification of secular terms in the second order set of equations yields the following slowly varying evolution equations:

$$\begin{aligned}
&2i\omega_s D_1 A + i\omega_s \hat{\delta}_1 A + 2k_{yy1}\overline{B}H_y Q e^{i\sigma_1 T_1} + k_{sw1} C H_s \overline{P} e^{i\sigma_2 T_1} = 0\\
&2i\omega_y D_1 B + i\omega_y \hat{\delta}_1 B + k_{sy2}\overline{A}H_y Q e^{i\sigma_1 T_1} = 0\\
&2i\omega_w D_1 C + i\omega_w \hat{\delta}_1 C + 2k_{ss3} A H_s P e^{-i\sigma_2 T_1} = 0\\
&2i\omega_p D_1 P - i\omega_p(\hat{\alpha}_p - \hat{\gamma}_p|P|^2)P - i\hat{\mu}_p\omega_p H_s P + (i\omega_q k_{p2} - \omega_q^2 k_{p3})Q^2 e^{i\sigma_p T_1} = 0\\
&2i\omega_q D_1 Q - i\omega_q(\hat{\alpha}_q - \hat{\gamma}_q|Q|^2)Q - i\hat{\mu}_q\omega_q H_y Q\\
&\quad -(i\omega_q k_{q2} - i\omega_p k_{q3} - \omega_p\omega_q k_{q4})P\overline{Q}e^{-i\sigma_p T_1} = 0\\
&2D_1 R - (\hat{\alpha}_r - \hat{\gamma}_r|R|^2)R - \hat{\mu}_r H_w R = 0.
\end{aligned} \tag{54}$$

Note that Eq. ((54)f) is decoupled from the rest of the equations and does not affect the estimation of γ_q. Therefore, we focus on the first five equations in Eq. (54). Substituting the polar form into Eq. (54) and separating real from imaginary parts yields ten evolution equations (see Appendix F). We identify the phase relations and reduce the evolution equations to the following five amplitude and three phase equations:

$$\begin{aligned}
D_1 a_s &= -\frac{1}{2}\hat{\delta}_1 a_s - \frac{1}{2\omega_s}k_{yy1}H_y a_y a_q \sin\phi_1 - \frac{1}{4\omega_s}k_{sw1}H_s a_w a_p \sin\phi_2\\
D_1 a_y &= -\frac{1}{2}\hat{\delta}_1 a_y - \frac{1}{4\omega_y}k_{sy2}H_y a_s a_q \sin\phi_1\\
D_1 a_w &= -\frac{1}{2}\hat{\delta}_1 a_w - \frac{1}{2\omega_w}k_{ss3}H_s a_s a_p \sin\phi_2\\
D_1 a_p &= \frac{1}{2}\hat{\alpha}_p a_p - \frac{1}{8}\hat{\gamma}_p a_p^3 + \frac{1}{2}\hat{\mu}_p H_s a_p - \frac{1}{4\omega_p}k_{p2}\omega_q a_q^2 \cos\phi_p + \frac{1}{4\omega_p}k_{p3}\omega_q^2 a_q^2 \sin\phi_p\\
D_1 a_q &= \frac{1}{2}\hat{\alpha}_q a_q - \frac{1}{8}\hat{\gamma}_q a_q^3 + \frac{1}{2}\hat{\mu}_q H_y a_q\\
&\quad + \frac{1}{4}k_{q2}a_p a_q \cos\phi_p - \frac{1}{4\omega_q}k_{q3}\omega_p a_p a_q \cos\phi_p + \frac{1}{4}k_{q4}\omega_p a_p a_q \sin\phi_p\\
D_1\phi_1 &= -\frac{1}{4}k_{q2}a_p \sin\phi_p + \frac{1}{4}k_{q3}\frac{\omega_p}{\omega_q}a_p \sin\phi_p + \frac{1}{4}k_{q4}\omega_p a_p \cos\phi_p\\
&\quad -\frac{1}{2}\frac{k_{yy1}}{\omega_s a_s}H_y a_y a_q \cos\phi_1 - \frac{1}{4}\frac{k_{sw1}}{\omega_s a_s}H_s a_w a_p \cos\phi_2 - \frac{1}{4}\frac{k_{sy2}}{\omega_y a_y}H_y a_s a_q \cos\phi_1 + \sigma_1\\
D_1\phi_2 &= \frac{1}{2}\frac{k_{ss3}}{\omega_w a_w}H_s a_s a_p \cos\phi_2 + \frac{1}{4}\frac{k_{p2}}{\omega_p a_p}\omega_q a_q^2 \sin\phi_p + \frac{1}{4}\frac{k_{p3}}{\omega_p a_p}\omega_q^2 a_q^2 \cos\phi_p\\
&\quad -\frac{1}{2}\frac{k_{yy1}}{\omega_s a_s}H_y a_y a_q \cos\phi_1 - \frac{1}{4}\frac{k_{sw1}}{\omega_s a_s}H_s a_w a_p \cos\phi_2 + \sigma_2\\
D_1\phi_p &= -\frac{1}{2}k_{q2}a_p \sin\phi_p + \frac{1}{2}k_{q3}\frac{\omega_p}{\omega_q}a_p \sin\phi_p + \frac{1}{2}k_{q4}\omega_p a_p \cos\phi_p\\
&\quad + \frac{1}{4}\frac{k_{p2}}{\omega_p a_p}\omega_q a_q^2 \sin\phi_p - \frac{1}{4}\frac{k_{p3}}{\omega_p a_p}\omega_q^2 a_q^2 \cos\phi_p + \sigma_p.
\end{aligned} \tag{55}$$

Setting the derivatives to zero, we obtain eight coupled algebraic equations, which after elimination of phase variables and some further manipulations (for details see Appendix F) yield five equations for the three unknown parameters k_{q2}, k_{q4}, γ (as two equations become redundant). We apply continuation analysis for the range of $13 < U^* < 20$ and the estimated coefficients and the numerical results are shown in Fig. 12. The maximal error in the 2:1 range is 3.58%.

It is instructive to compare the computed normalized transverse limit-cycle frequencies to the measured data. The result in Fig. 13 shows that in the region where the 2:1 internal resonance is valid, the computed $\omega_{y,\text{LC}}$ is comparable to the measured f^* which overcomes the discrepancy in Fig. 11 beyond the valid range of the 1:1 assumption.

4.3. The bifurcation structure

In this section, we show the time series, state space projections and Poincaré maps at various Reynolds numbers and for different modes of internal resonance for the tethered sphere obtained numerically from the complete model Eq. (7). The Poincaré maps (shown in blue dots) are sampled every time the transverse sphere velocity y_τ or the transverse wake

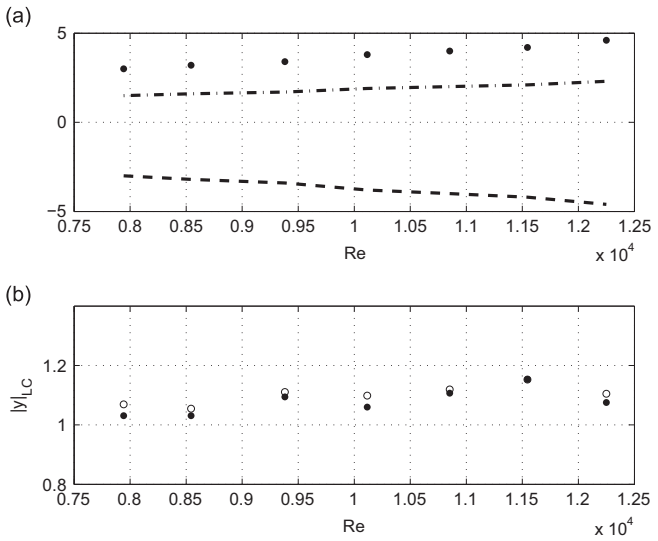


Fig. 12. (a). The estimated γ_q and wake coupling coefficients. - - k_{q2}, -.- k_{q3}, •γ. (b) The comparison between the numerical and measured transverse limit-cycle amplitude in the range that is valid for the assumption $\omega_y:\omega_q = 1:2$. ○ measured, • numerical results with parameters estimated from the 1:1:5:4:2: ω_r model.

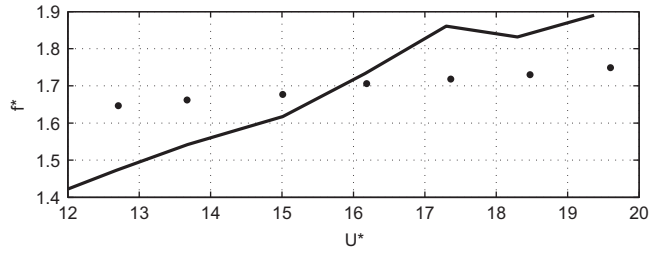


Fig. 13. The measured normalized frequency f^* in Govardhan and Williamson (1997) • the numerically obtained normalized transverse limit-cycle frequency $\omega_{y,LC}/\omega_y$ with internal resonance $\omega_q:\omega_y = 2:1$.

velocity q_τ has an upward zero-crossing (Nayfeh and Balachandran, 1996). The bifurcation structure combining the 1:1 and 2:1 internal resonance results for the complete range ($2000 < U^* < 13\,000$) of the experiment in Govardhan and Williamson (1997) is depicted in Fig. 14. Note that there are two distinct regions: (i) $0 < \mathrm{Re} < 2450$ stationary dynamics at steady state and (ii) $2450 < \mathrm{Re} < 13\,000$ VIV where the system motion is periodic. Note that the maximal difference between the measured and predicted limit-cycle amplitude $|y|_{LC}$ is 6.29%.

Two selected examples of sphere and wake dynamics are shown for $\mathrm{Re} = 6000$ and $\mathrm{Re} = 12\,250$ in Fig. 15. Note that as expected the period of the transverse wake q is double that of the streamwise wake p. Furthermore, recall that the structural resonance frequencies in the horizontal plane are equal ($\omega_x = \omega_y$ in Eq. (15)). However according to the internal resonance prediction in Section 4.2.1, the limit-cycle periods for x, y are identical to those of the corresponding wakes p, q.

The dynamical system response is shown in the state space projections of the sphere $y(x)$ and wake $q(p)$ for $\mathrm{Re} = 6000$ and $\mathrm{Re} = 12\ \ 250$ in Fig. 16. Note that the relationship between the periods in Fig. 15 is depicted by the classical figure eight in Fig. 16, which corresponds to that of Fig. 2 in Govardhan and Williamson (1997). Furthermore, while the single Poincaré point in Fig. 16 depicts the periodic nature of the response, the increase in upstream velocity from $\mathrm{Re} = 6000$ to $\mathrm{Re} = 12\ \ 250$ reveals a secondary internal loop in the $y(x)$ projection in Fig. 16(b).

We construct the physical three dimensional state space projections $z(x, y)$ for the two selected examples in Fig. 17, and note that both, while distorted, preserve the classical figure eight form demonstrating symmetry about $y = 0$.

The key points resulting from the combined asymptotic and numerical analyses include: (i) the cubic wake parameter γ_q can readily be estimated from the reduced-order analysis; (ii) the quadratic wake coupling coefficients are estimated from the full model for two separate internal resonance conditions identified by the ratio of the limit-cycle frequency ω_y and its corresponding wake frequency ω_q. Specifically $\omega_q:\omega_y = 1:1$ for $3 < U^* < 13$ and $\omega_q:\omega_y = 2:1$ for $13 < U^* < 20$. We note that this parameter estimation methodology is not unique but can be validated by comparison of the numerically obtained limit-cycle amplitudes and frequencies with the measurements in the respective ranges of internal resonance conditions. The figure eight topology in the $y(x)$ state space projection is preserved throughout the range of the Govardhan and Williamson (1997) experiment ($3 < U^* < 20$).

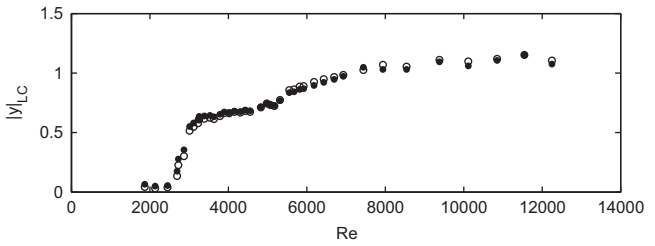


Fig. 14. Bifurcation diagram of the transverse limit-cycle amplitude $|y|_{LC}$ vs. Re: ● the numerical results, ○ the measured data in Govardhan and Williamson (1997).

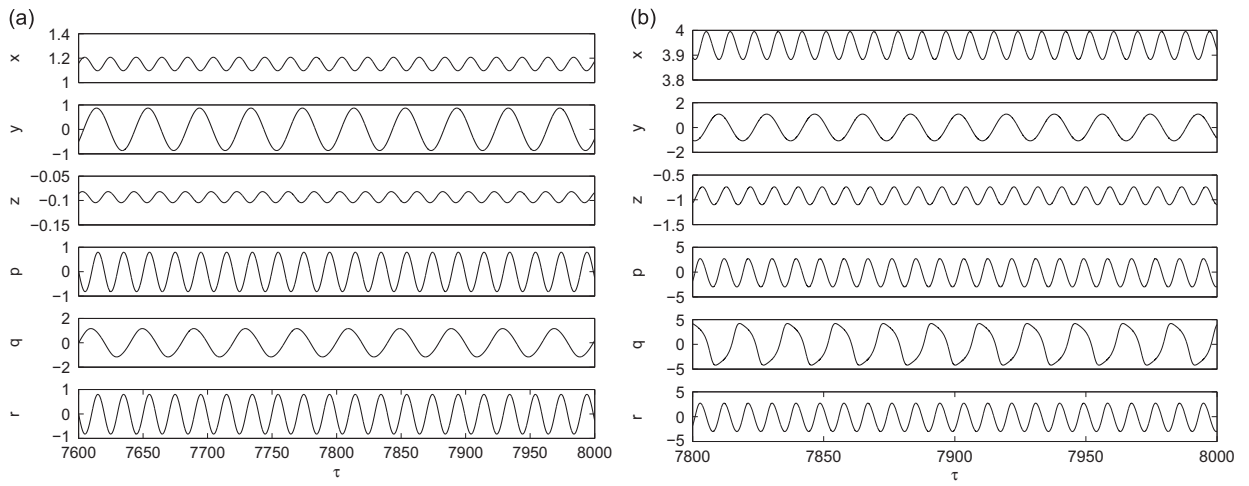


Fig. 15. The steady state time series at (a) Re=6000, $\omega_q:\omega_y=1:1$ and (b) Re=12 250, $\omega_q:\omega_y=2:1$.

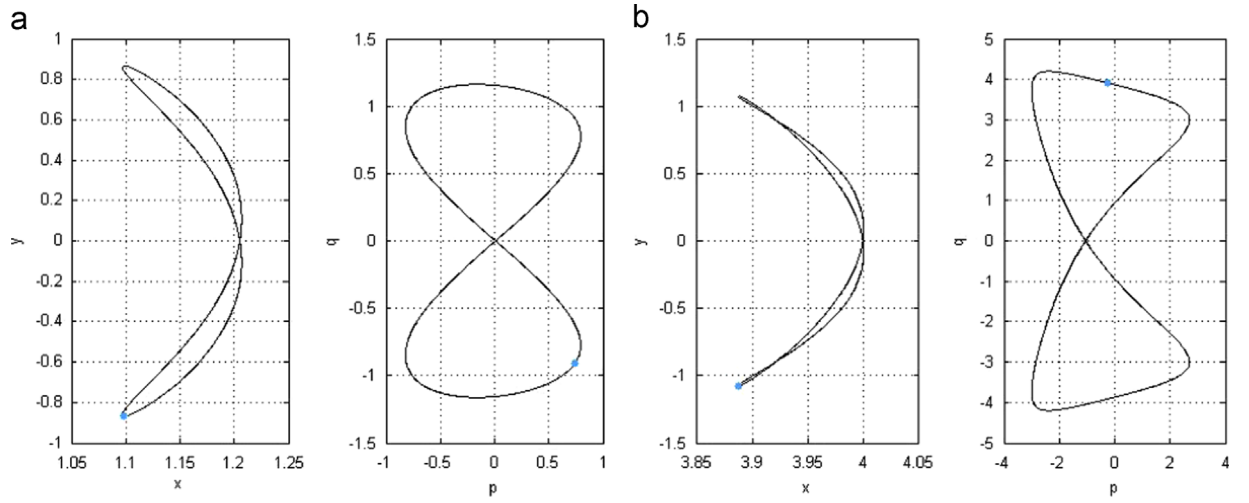


Fig. 16. The steady state space projections $y(x)$ and $q(p)$ at (a) Re=6000, $\omega_q:\omega_y=1:1$ and (b) Re=12 250, $\omega_q:\omega_y=2:1$.

5. Numerical analysis of a tethered spherical aerostat

5.1. Model-based estimation

In this section, we will adapt the model validated for the tethered sphere to that of a tethered aerostat investigated by Coulomb-Pontbriand and Nahon (2009). The system parameters which are close to those of the sphere in Govardhan and Williamson (1997) for the aerostat system are $M^*=0.49$, $L^*=8.55$ for the first case and $L^*=12.83$ for the second case, resulting in a nondimensional inverse added mass parameter $\eta=0.0886$ and $\eta=0.059$ respectively. As noted by Coulomb-Pontbriand and Nahon (2009) the aerostat experiments for $L^*=8.55$ and 12.83 were conducted for larger range of reduced

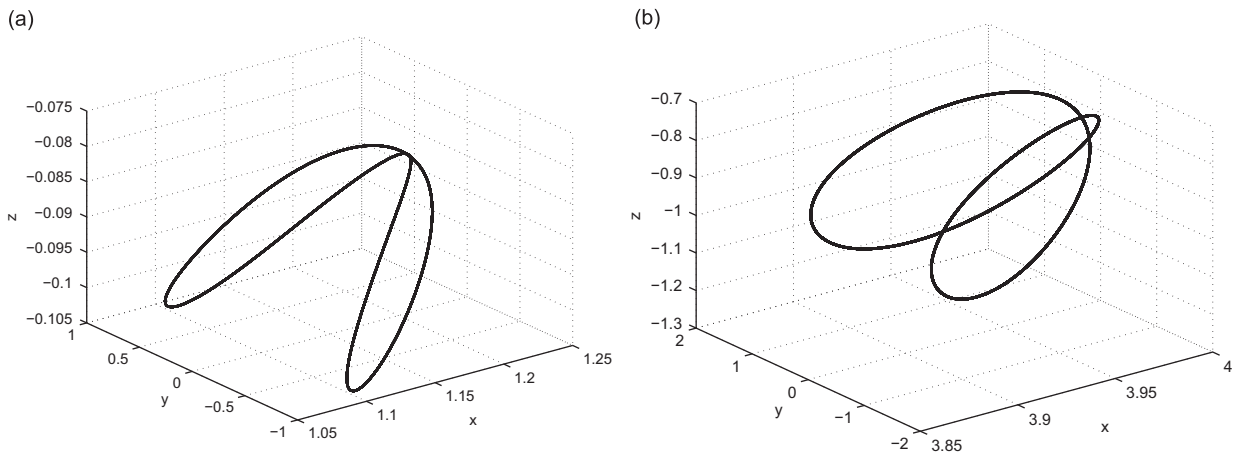


Fig. 17. The physical three dimensional state space projection $z(x,y)$ at (a) Re=6000, $\omega_q:\omega_y=1:1$ and (b) Re=12 250, $\omega_q:\omega_y=2:1$.

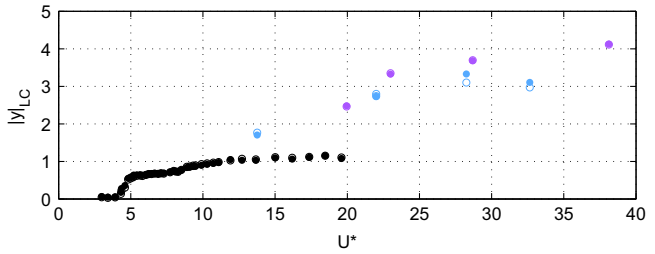


Fig. 18. The transverse limit-cycle amplitudes. ○ measured data, ● numerical results. Black indicates Govardhan and Williamson (1997) experiment. Blue indicates Coulomb-Pontbriand and Nahon (2009) experiment with $L^*=8.55$. Purple indicates Coulomb-Pontbriand and Nahon (2009) experiment with $L^*=12.83$. (For interpretation of the references to color in this figure caption, the reader is referred to the web version of this article.)

velocities ($20<U^*<40$) than those of Govardhan and Williamson (1997) for $L^*=8.93$ ($4<U^*<20$). However, as the average mean drag of the largest reduced velocity in Govardhan and Williamson (1997) (eg. Fig. 2(a) for Re > 10 000) is $\beta_D=0.68$, which is within the range of the estimated mean drag of Coulomb-Pontbriand and Nahon (2009) (see Fig. 6 in Coulomb-Pontbriand and Nahon, 2009), we make use of this value and the average pretension of $\alpha=0.97$ throughout this section. Considering the different experimental condition of Coulomb-Pontbriand and Nahon (2009) which is conducted in air, instead of in water as Govardhan and Williamson (1997), the linear damping ratio ζ is reduced from $\zeta_{GW97}=0.08$ to $\zeta_{Nahon09}=0.008$, corresponding to damping parameter $\delta=2.69e-3$ for $L^*=8.55$ and $\delta=4.89e-4$ for $L^*=12.83$ respectively. Recall that the combination of the linear van der Pol coefficient α_q and the coupling coefficient μ_q is obtained by the Hopf condition. However, as the experimental conditions of Coulomb-Pontbriand and Nahon (2009) were beyond the Hopf threshold, we also make use of the values estimated in Section 3 for the conditions of Govardhan and Williamson (1997) ($\alpha_q=7e-4$ and $\mu_q=2.62e-2$).

In the validation for of the tethered sphere model, the derivation of the nonlinear van der Pol coefficient γ_q and the wake coupling coefficients were based on asymptotic analysis of 1:1 and 2:1 internal resonance conditions. However for the case of the spherical aerostat (Coulomb-Pontbriand and Nahon, 2009), the response to non-stationary wind excitation did not yield a distinct limit-cycle frequency (or f^*). Thus, we use the estimates based on Govardhan and Williamson (1997) for ω_q and ω_y. Note that the case where $L^*=8.55$ is similar to $L^*=8.93$ in Govardhan and Williamson (1997) supports the assumptions that the natural frequencies are similar (see Fig. 6(a)) and that conditions for internal resonance include 2:1 and 3:1 in the range $20<U^*<40$ (see Fig. 6(b)). The nonlinear parameters are thus determined by matching the documented amplitude in the numerical simulation. The numerical and measured data for Coulomb-Pontbriand and Nahon (2009) are shown in Fig. 18 overlaid with the previous results for Govardhan and Williamson (1997).

For the case of $L^*=8.55$, for $U^*=22$ computed with a 2:1 internal resonance assumption, the aerostat vibrates periodically as can be seen in the time series and the state space projections overlaid with the single Poincaré point in Fig. 19.

For the case of $U^*=28$, we assume a 3:1 internal resonance based on the extrapolated frequency behavior in Fig. 6. Fig. 20 depicts the time series and statespace projections of $U^*=28$ which appear quasiperiodic.

In order to confirm the nature of the quasiperiodic response, we plot the Poincaré maps in Fig. 21, and note that both the $y(x)$ and $q(p)$ projections portray torus-like behavior where the Poincaré points define a distinct topological closed curve. Note that the Poincaré map in the $y(x)$ projection covers the complete state space (eg. $-3.2<y<3.2$) whereas the map for the $q(p)$ projection is localized to the lower part of the state space. These maps are similar to those of the chaotic string,

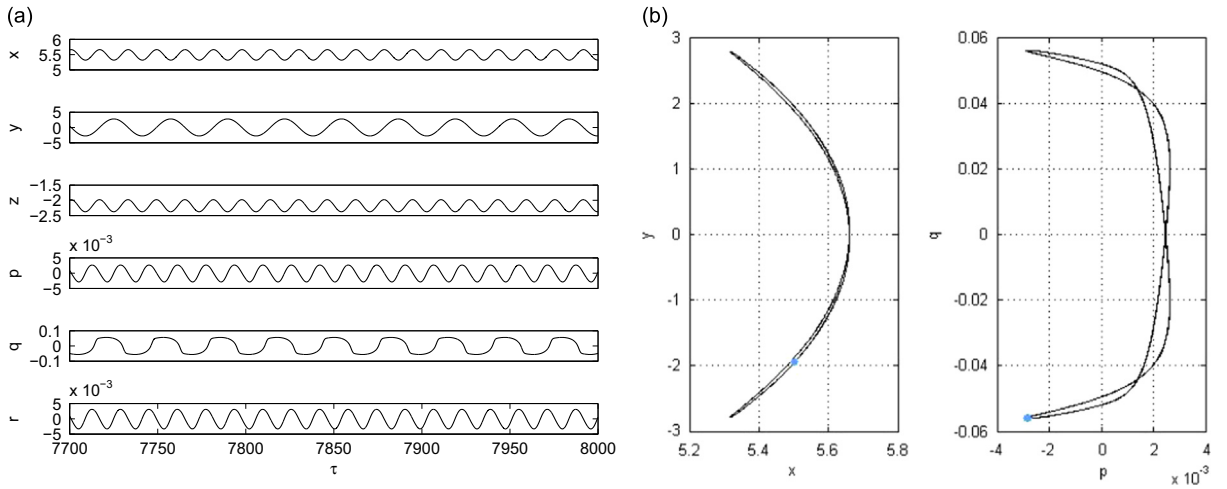


Fig. 19. (a) The time series and (b) the state spaces at $U^*=22$ for $L^*=8.55$. $\omega_q:\omega_q=2:1$.

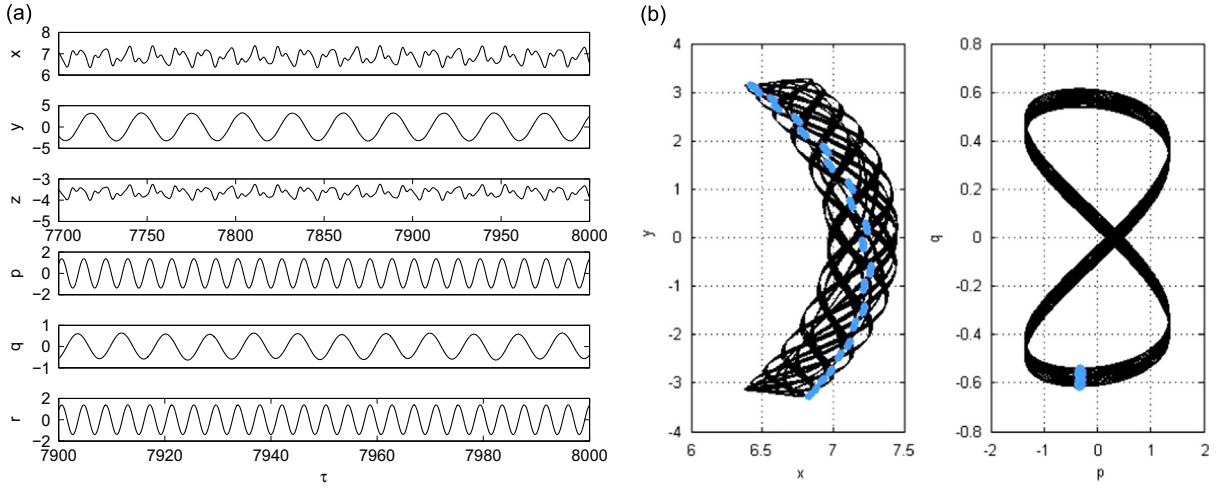


Fig. 20. (a) The time series and (b) the state spaces at $U^*=28$ for $L^*=8.55$. $\omega_q:\omega_q=3:1$.

where complete coverage was defined as transit and a localized map was defined as non-transit (Rubin and Gottlieb, 1996; O'Reilly and Holmes, 1992).

The case of $U^*=32$ is depicted via the time series and Poincaré maps in Fig. 22. Note that the Poincaré map for $y(x)$ defines a transit quasiperiodic torus whereas the Poincaré map for $q(p)$ defines a non-transit cloud which has no topological structure. In the following section we will demonstrate that this unstructured behavior leads to a chaotic-like torus for $U^*=31$.

For the case of $L^*=12.83$, a 2:1 internal resonance is assumed in the range $20<U^*<29$. A typical solution in this interval is depicted in the time series for $U^*=29$ of Fig. 23(a). While the time series appears periodic, a close observation of the Poincaré map in Fig. 23(b) reveals its non-transit quasiperiodic nature.

The region in the range $29<U^*<40$ is assumed to be that of a 3:1 internal resonance. A typical solution is depicted in Fig. 24 which shows that the response is transit quasiperiodic.

5.2. The bifurcation structure

Based on the numerical results presented in Section 5.1, we conduct a numerical continuation analysis within each internal resonance mode. The bifurcation diagram for the case of $L^*=8.55$ is portrayed in Fig. 25.

We assume a 1:1 internal resonance for the low reduced velocities. The aerostat motion remains periodic until $U^*=13$, when quasiperiodic motion begins to appear with Poincaré maps similar to that of Fig. 28. The quasiperiodic motion continues to $U^*=16$ where the 1:1 assumption no longer holds. Therefore for $15<U^*<26$ we assume a 2:1 internal resonance. In this interval, the aerostat motion is generally weakly quasiperiodic, the topology is similar to that of Fig. 23(b).

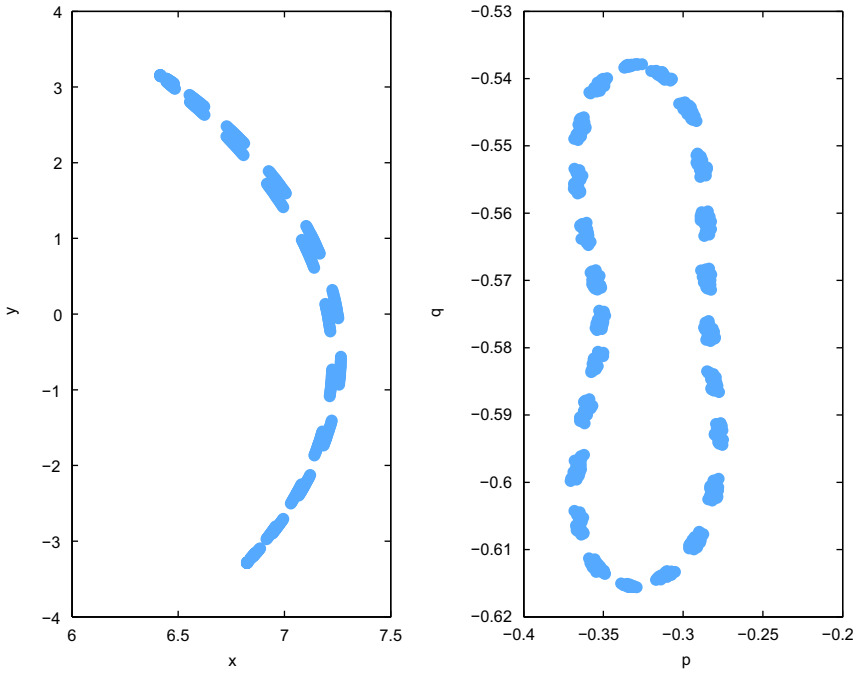


Fig. 21. The Poincaré maps at $U^*=28$ for $L^*=8.55$. $\omega_q:\omega_y=3:1$.

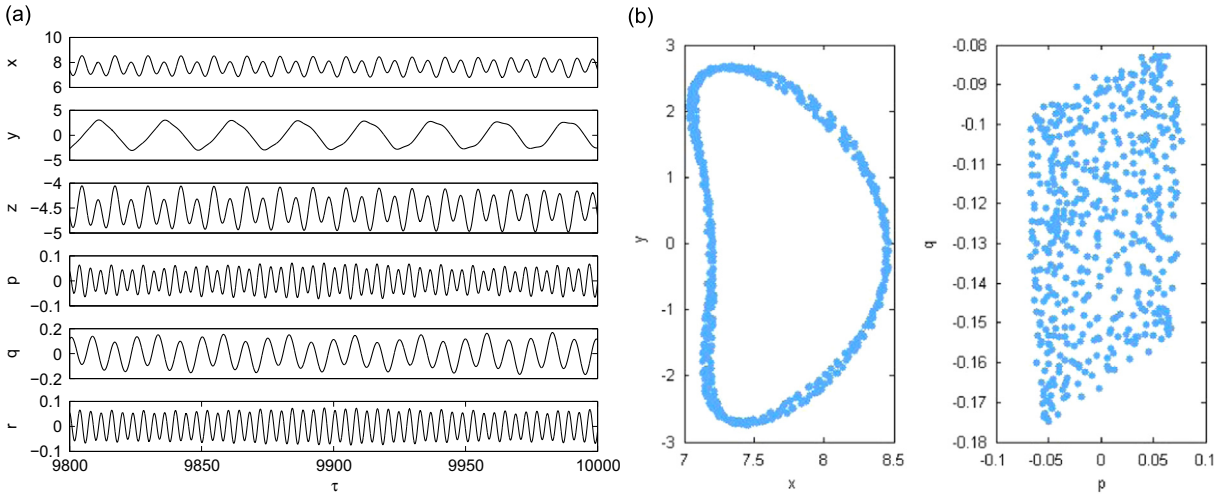


Fig. 22. (a) The time series and (b) the Poincaré maps at $U^*=32$ for $L^*=8.55$. $\omega_q:\omega_y=3:1$.

From $U^*>27$ we assume a 3:1 internal resonance. The system first exhibits quasiperiodic motions ($27\leq U^*<30$) similar to that in Fig. 21, before it exhibits non-stationary chaotic-like motions ($31\leq U^*\leq 32$) shown in Fig. 22, and Fig. 26.

The three dimensional state space projections for $U^*=22,28,31$ are portrayed in Fig. 27. We note that the periodic response ($U^*=22$) has the figure eight form of the tethered sphere (Govardhan and Williamson, 1997) in Fig. 17. Furthermore, both quasiperiodic ($U^*=28$) and chaotic ($U^*=31$) projections appear to evolve about a single inclined plane similar to that of the periodic limit-cycles.

5.3. Sensitivity of parameters and amplitude reduction

We note that the bifurcation analysis revealed several possible solutions obtained by changing the internal resonance conditions. An example is the transition between the 1:1 and 2:1 internal resonances. Fig. 28(a) and (b) depict two solutions for $U^*=14$ obtained using 1:1 and 2:1 conditions respectively. Note that they exhibit existence of a quasiperiodic torus (Fig. 28(a)) and a periodic limit-cycle (Fig. 28(b)).

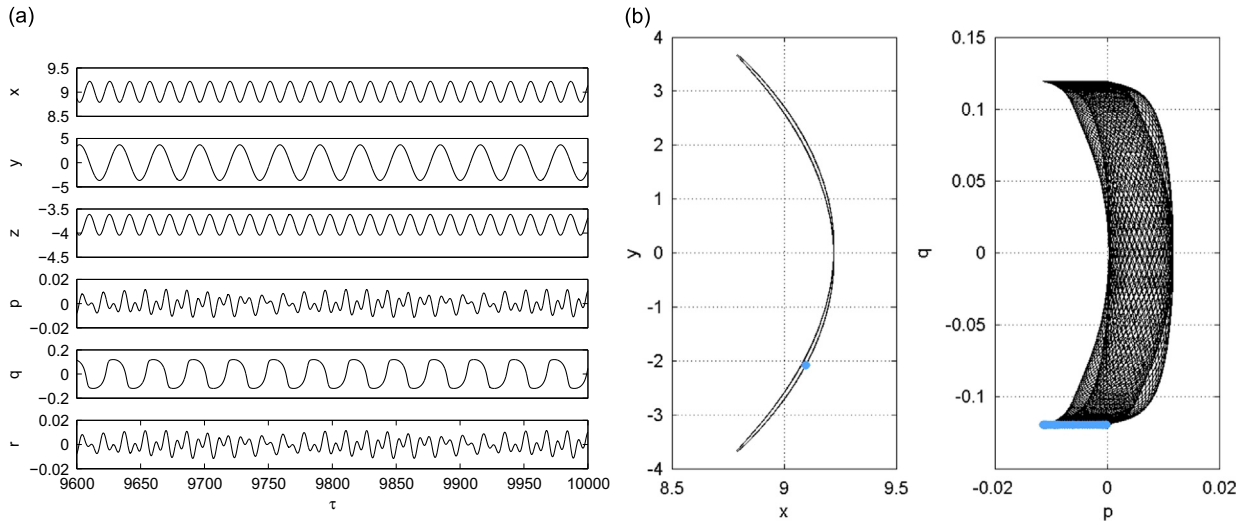


Fig. 23. (a) The time series and (b) the state spaces at $U^*=29$ for $L^*=12.83$. $\omega_q:\omega_y=2:1$.

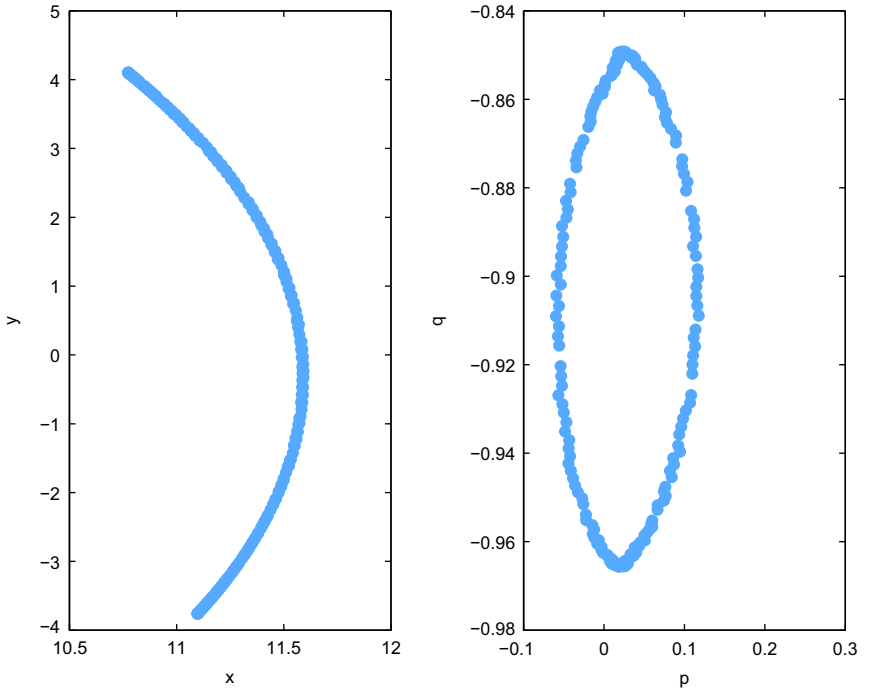


Fig. 24. The Poincaré maps at $U^*=38$ for $L^*=12.83$. $\omega_q:\omega_y=3:1$.

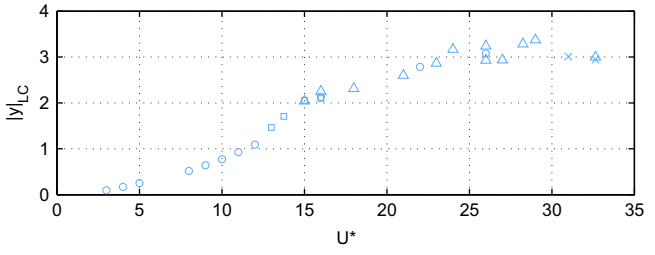


Fig. 25. The bifurcation diagram for $L^*=8.55$. ○ periodic, ▵ quasiperiodic with one torus, □ quasiperiodic with two tori, × chaotic like.

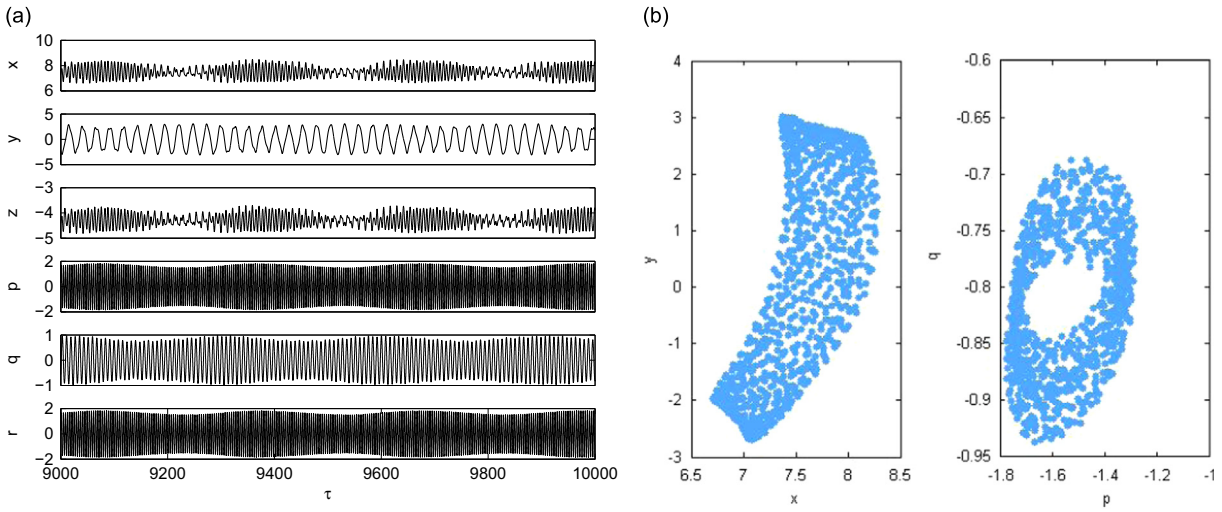


Fig. 26. (a) The time series and (b) the Poincaré maps at $U^*=31$. $\omega_q:\omega_y=3:1$.

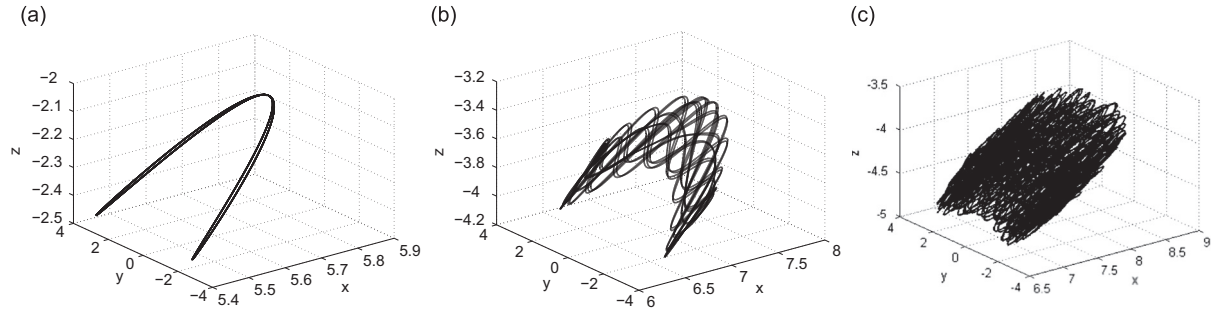


Fig. 27. The three dimensional state space projections for (a) $U^*=22$, (b) $U^*=28$ and (c) $U^*=31$.

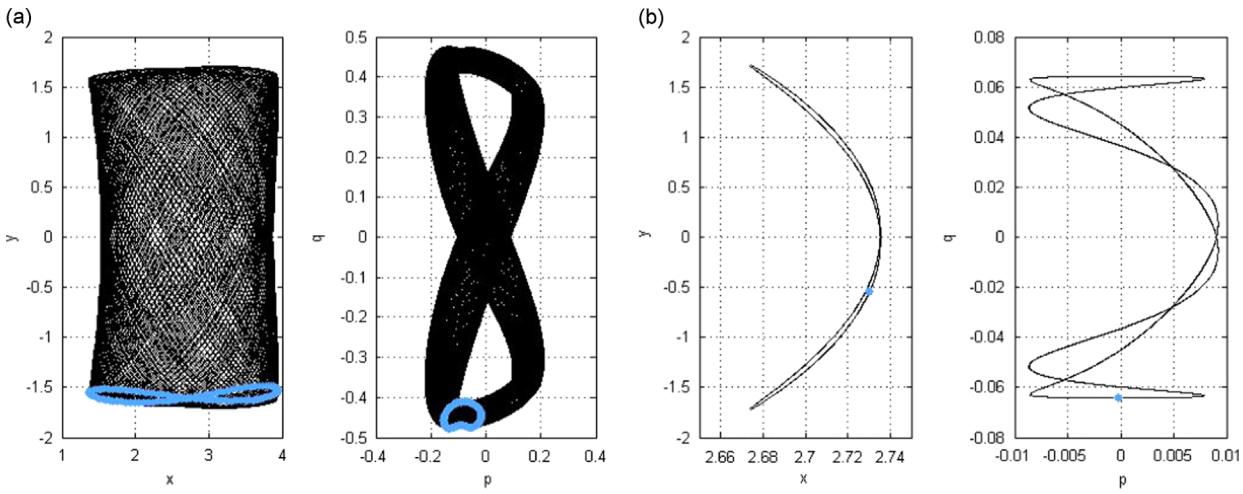


Fig. 28. The state spaces at $U^*=14$. (a) $\omega_q:\omega_y=1:1$ and (b) $\omega_q:\omega_y=2:1$.

For $U^*=26$, a weakly quasiperiodic solution similar to that in Fig. 23 was obtained with a 2:1 internal resonance. However use of a 3:1 internal resonance condition shows chaotic-like motions for $U^*=26$ in Fig. 29.

The different solutions at the transitions demonstrate the sensitivity of parameters in this model-based estimation procedure. We note that additional experimental data (e.g. the limit-cycle frequency f^*) would enable a more precise conclusion on which condition is accurate.

The proposed model provides several mechanisms for limit-cycle amplitude reduction. We consider here an increase in the linear structural damping parameter and show that the response for $U^*=28$ and $U^*=31$ can decrease by 50.7% in

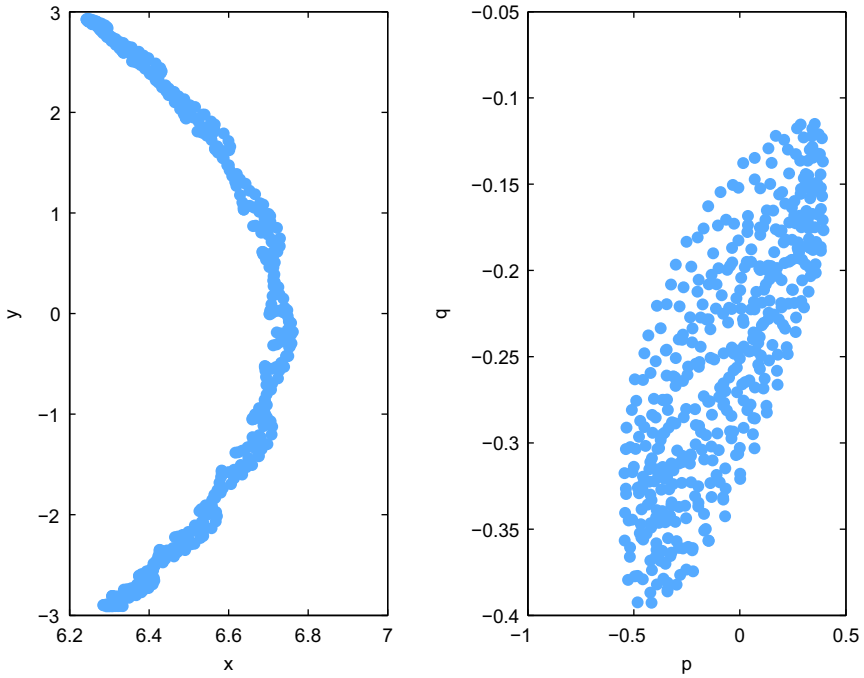


Fig. 29. The Poincaré maps at $U^*=26$. $\omega_q:\omega_y=3:1$.

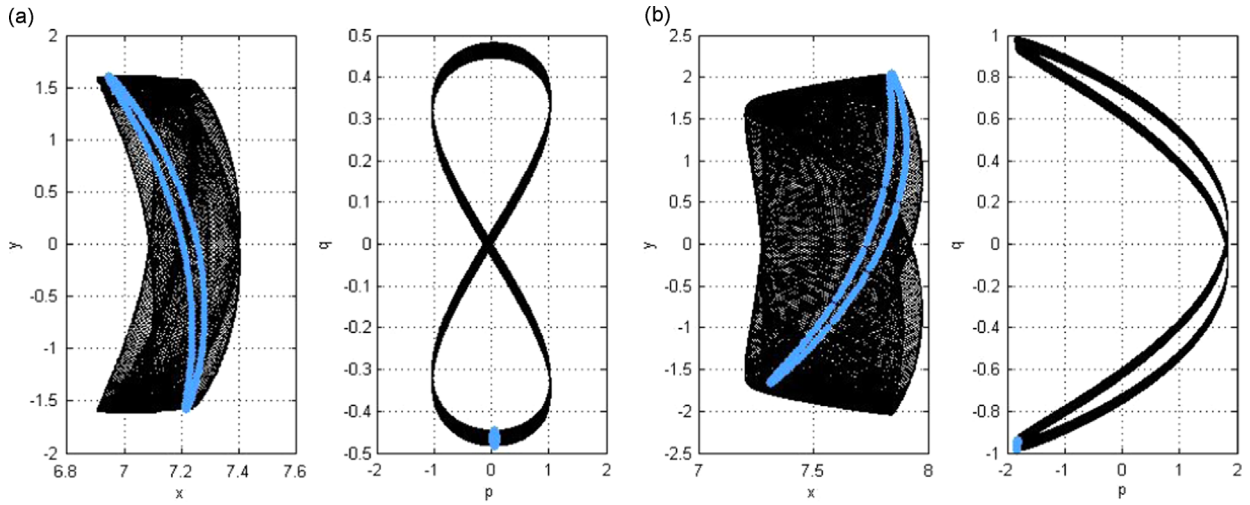


Fig. 30. The state space projections with decreased limit-cycle amplitude for increased structural damping: (a) $L^*=8.55$, $U^*=28$, $\delta=2.83e-3$, (b) $L^*=8.55$, $U^*=31$, $\delta=3.16e-3$.

Fig. 30(a) and 31.8% in Fig. 30(b) respectively. We note that the response for $U^*=28$ was originally quasiperiodic (Fig. 21) and only reduced in amplitude (see Fig. 30(a)). However, the response for $U^*=31$ changed its topology from chaotic-like behavior (Fig. 26) and in addition to amplitude reduction, became quasiperiodic (see Fig. 30(b)).

6. Closing remarks

We derive and investigate the nonlinear dynamics of a tethered sphere. A Lagrangian based model augmented by wake oscillators for the streamwise, vertical and transverse directions enables model-based estimation of both structural and aeroelastic parameters via asymptotic analysis of internal resonance conditions between the transverse wake frequency and its structural counterpart. Validation of the proposed methodology is demonstrated by comparison of results with those of a benchmark experiment for a light sphere in water conducted by Govardhan and Williamson (1997) and by analysis of a tethered aerostat experiment conducted by Coulomb-Pontbriand and Nahon (2009). A numerical bifurcation analysis of the

validated aerostat system reveals existence of possible quasiperiodic and non-stationary chaotic-like solutions that are consistent with documented instabilities of aerostats in severe environmental conditions. We note that the bifurcation analysis also revealed that the model parameter selection is sensitive particularly at the transitions from one resonance condition to another. This deficiency can be overcome with additional experimental data that will enable a more precise estimation procedure.

We summarize our model-based estimation procedure as follows: (*i*) derivation of a coupled fluid–structure interaction dynamical system that consistently includes both structural nonlinearities and the coupling with the three-dimensional time-dependent fluctuating wake components; (*ii*) estimation of the linear parameter set can be systematically done by considering the equilibrium position and its stability near the VIV threshold; (*iii*) the nonlinear fluctuating wake parameters can be obtained by multiple-scale asymptotics near selected internal resonance conditions; (*iv*) model accuracy should be investigated by comparison with documented experiments or detailed solution from direct numerical simulations of high-order continuum fluid mechanics models; and (*v*) a comprehensive bifurcation analysis of the validated system can reveal its intricate nature and shed light on the transitions from periodic to non-stationary dynamics and further enable consideration and optimization of different limit-cycle amplitude suppression mechanisms.

We close with a remark that the proposed framework can be extended to incorporate rotations of asymmetrical bluff rigid-bodies and the continuum spatio-temporal behavior of cable dynamics. Both aerostat and cable dynamics will thus exhibit additional non-negligible effects such as internal and combination resonances influencing the fluctuating streamwise, lift and transverse wakes that govern the response dynamics and its bifurcation structure.

Acknowledgments

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Appendix A. Coefficients for equilibrium of the reduced-order model evolution equations

The two algebraic equations expressed in terms of a_q and γ_q derived from the equilibrium of the slowly varying evolution equations obtained from Eq. (35) are

$$c_1\hat{\gamma}_q^2 a_q^8 + c_2\hat{\gamma}_q a_q^6 + c_3 a_q^4 + c_4 a_q^2 + c_5 = 0$$
$$d_1\hat{\gamma}_q a_q^6 + d_2 a_q^4 + d_3 a_q^2 + d_4 = 0, \tag{56a, b}$$

where

$$c_1 = (\hat{\eta}u^2\omega_q)^2,\ c_2 = -8(\hat{\eta}u^2\omega_q)^2\hat{\alpha}_q,\ c_3 = 16(\hat{\eta}u^2\hat{\alpha}_q\omega_q)^2,$$
$$c_4 = -16(\hat{\eta}u^2\hat{\mu}_q\omega_y a_y)^2,\ c_5 = 16(\hat{\delta}_1\hat{\mu}_q)^2(\omega_y a_y)^4,$$
$$d_1 = -(\hat{\eta}u^2\omega_q)^2,\ d_2 = 4(\hat{\eta}u^2\omega_q)^2\hat{\alpha}_q,$$
$$d_3 = 3\hat{\eta}u^2\tilde{\alpha}\hat{\mu}_q\omega_q\omega_y a_y^4 - 8\hat{\eta}u^2\hat{\mu}_q\sigma_y\omega_q\omega_y^2 a_y^2,\ d_4 = 4\hat{\delta}_1\hat{\mu}_q^2(\omega_y a_y)^4. \tag{57}$$

Appendix B. The conservative restoring force terms to cubic order

We translate the system in Eq. (7) to coordinates about the equilibrium position ($s = x - x^*, y, w = z - z^*$) and expand the conservative restoring force to cubic order

$$f_s = R^*x^* + \omega_s^2 s + k_{w1}w + k_{ss1}s^2 + k_{yy1}y^2 + k_{ww1}w^2 + k_{sw1}sw$$
$$f_y = \omega_y^2 y + k_{sy2}sy + k_{yw2}yw$$
$$f_w = R^*(1+z^*) + \omega_w^2 w + k_{s3}s + k_{ss3}s^2 + k_{yy3}y^2 + k_{ww3}w^2 + k_{sw3}sw, \tag{58}$$

where

$$k_{s1} = -a_{21},\quad k_{w1} = -a_{25},\quad k_{s3} = k_{w1},\quad k_{w3} = -a_{65},$$
$$k_{ss1} = \frac{3\alpha x^*(1+z^*)^2}{2(x^{*2}+(1+z^*)^2)^{5/2}},\quad k_{yy1} = \frac{\alpha x^*}{2(x^{*2}+(1+z^*)^2)^{3/2}},$$
$$k_{ww1} = \frac{\alpha x^*(x^{*2}-2(1+z^*)^2)}{2(x^{*2}+(1+z^*)^2)^{5/2}},\quad k_{sw1} = -\frac{\alpha(1+z^*)(2x^{*2}-(1+z^*)^2)}{(x^{*2}+(1+z^*)^2)^{5/2}},$$
$$k_{sy2} = \frac{\alpha x^*}{(x^{*2}+(1+z^*)^2)^{3/2}},\quad k_{yw2} = \frac{\alpha(1+z^*)}{(x^{*2}+(1+z^*)^2)^{3/2}},\quad k_{ss3} = \frac{1}{2}k_{sw1},$$

$$k_{yy3}=\frac{\alpha(1+z^*)}{2(x^{*2}+(1+z^*)^2)^{3/2}},\quad k_{ww3}=\frac{3\alpha x^{*2}(1+z^*)}{2(x^{*2}+(1+z^*)^2)^{5/2}},\quad k_{sw3}=2k_{ww1}. \tag{59}$$

Note that the constant terms will cancel the gravity and the hydrodynamic bias.

Investigation of the coefficients in the range $2000<\mathrm{Re}<12\,250$ reveals that three coefficients k_{w1},k_{s3},k_{ss1} linearly increase from 0 to 0.5. Whereas the rest of the coefficients remain constant throughout the range ($k_{yy1}=0.2228$, $k_{ww1}=-0.3922$, $k_{sw1}=0.2872$, $k_{sy2}=0.4455$, $k_{yw2}=0.8404$, $k_{ss3}=0.1436$, $k_{yy3}=0.4202$, $k_{ww3}=0.2766$, $k_{sw3}=-0.7844$).

Appendix C. Evolution equations and limit-cycle amplitude equations for the 1:1:6:2:1:1 internal resonance

The six slowly varying evolution equations for the amplitudes a_y, a_p, a_q and angles $\theta_y,\theta_p,\theta_q$ are

$$\begin{aligned}
D_1a_y&=-\frac{1}{2}\hat{\delta}_1a_y+\frac{1}{2}\frac{\hat{\eta}}{\omega_y}u^2a_q\sin\phi_y\\
D_1\theta_y&=-\frac{1}{2}\frac{\hat{\eta}}{\omega_y}u^2\frac{a_q}{a_y}\cos\phi_y\\
D_1a_p&=\frac{1}{2}\hat{\alpha}_pa_p-\frac{1}{8}\hat{\gamma}_pa_p^3-\frac{\omega_qa_q^2}{4\omega_p}\left(k_{p2}\cos\phi_p-\omega_qk_{p3}\sin\phi_p\right)\\
D_1\theta_p&=-\frac{\omega_qa_q^2}{4\omega_pa_p}\left(k_{p2}\sin\phi_p+\omega_qk_{p3}\cos\phi_p\right)\\
D_1a_q&=\frac{1}{2}\hat{\alpha}_qa_q-\frac{1}{8}\hat{\gamma}_qa_q^3+\frac{1}{2}\frac{\omega_y}{\omega_q}\hat{\mu}_qa_y\cos\phi_y\\
&\quad-\frac{1}{4}\left(\frac{\omega_p}{\omega_q}k_{q3}-k_{q2}\right)a_pa_q\cos\phi_p+\frac{1}{4}\omega_pk_{q4}a_pa_q\sin\phi_p\\
D_1\theta_q&=\frac{1}{4}\left(\frac{\omega_p}{\omega_q}k_{q3}-k_{q2}\right)a_p\sin\phi_p+\frac{1}{4}\omega_pk_{q4}a_p\cos\phi_p-\frac{1}{2}\frac{\omega_y}{\omega_q}\hat{\mu}_q\frac{a_y}{a_q}\sin\phi_y,
\end{aligned} \tag{60}$$

where $\phi_y=\theta_q-\theta_y+\sigma_yT_1$, $\phi_p=2\theta_q-\theta_p+\sigma_pT_1$. The equilibrium of Eq. (48) reads

$$\begin{aligned}
&\omega_y\hat{\delta}_1a_y-\hat{\eta}u^2a_q\sin\phi_y=0\\
&4\omega_p\hat{\alpha}_pa_p-\omega_p\hat{\gamma}a_p^3-2\omega_qa_q^2k_{p2}\cos\phi_p+2\omega_q^2a_q^2k_{p3}\sin\phi_p=0\\
&4\omega_q\hat{\alpha}_qa_q-\omega_q\hat{\gamma}a_q^3+4\omega_y\hat{\mu}_qa_y\cos\phi_y-2\omega_pa_pa_qk_{q3}\cos\phi_p+2\omega_qa_pa_qk_{q2}\cos\phi_p\\
&\quad+2\omega_p\omega_qa_pa_qk_{q4}\sin\phi_p=0\\
&\omega_y\omega_pa_ya_pa_qk_{q3}\sin\phi_p-\omega_y\omega_qa_ya_pa_qk_{q2}\sin\phi_p+\omega_y\omega_p\omega_qa_ya_pa_qk_{q4}\cos\phi_p\\
&\quad-2\omega_y^2\hat{\mu}_qa_y^2\sin\phi_y+2\omega_q\hat{\eta}u^2a_q^2\cos\phi_y+4\omega_y\omega_q\sigma_ya_ya_q=0\\
&2\omega_p^2a_p^2a_qk_{q3}\sin\phi_p-2\omega_p\omega_qa_p^2a_qk_{q2}\sin\phi_p+2\omega_p^2\omega_qa_p^2a_qk_{q4}\cos\phi_p\\
&\quad-4\omega_y\omega_p\hat{\mu}_qa_ya_p\sin\phi_y+\omega_q^2a_q^3k_{p2}\sin\phi_p+\omega_q^3a_q^3k_{p3}\cos\phi_p+4\omega_p\omega_q\sigma_pa_pa_q=0.
\end{aligned} \tag{61}$$

Appendix D. Coefficients of the algebraic equations in Eq. (49)

The coefficients are expressed in original parameters with the assumptions that $\sigma_y=\sigma_p=0$, $\hat{\gamma}_p=\hat{\gamma}_q=\hat{\gamma}$ and $\hat{\alpha}_p=\hat{\alpha}_q$

$$\begin{aligned}
m_{11}&=(4\hat{\alpha}_p-\hat{\gamma}a_p^2)^2\omega_p^2\omega_q^2a_p^4-4\omega_q^6a_p^2a_q^4\\
m_{12}&=(4\hat{\alpha}_p-\hat{\gamma}a_p^2)^2\omega_p^4a_p^4-4\omega_p^2\omega_q^4a_p^2a_q^4\\
m_{13}&=(4\hat{\alpha}_p-\hat{\gamma}a_p^2)^2\omega_p^4\omega_q^2a_p^4-4\omega_p^2\omega_q^4a_p^2a_q^4\\
m_{14}&=-2(4\hat{\alpha}_p-\hat{\gamma}a_p^2)^2\omega_p^3\omega_qa_p^4+8\omega_p\omega_q^5a_p^2a_q^4\\
m_{15}&=-8\omega_p\omega_q^5a_p^2a_q^4\\
m_{16}&=8\omega_p^2\omega_q^4a_p^2a_q^4\\
m_{17}&=2\omega_p\omega_q^2(4\hat{\alpha}_p-\hat{\gamma}a_p^2)((4\alpha_q-\hat{\gamma}a_q^2)\omega_qa_q+4\hat{\mu}_q\omega_ya_y\cos\phi_y)a_p^2a_q\\
m_{18}&=-2\omega_p^2\omega_q(4\hat{\alpha}_p-\hat{\gamma}a_p^2)((4\alpha_q-\hat{\gamma}a_q^2)\omega_qa_q+4\hat{\mu}_q\omega_ya_y\cos\phi_y)a_p^2a_q\\
m_{19}&=-2\omega_p^2\omega_q^3(4\hat{\alpha}_p-\hat{\gamma}a_p^2)((4\alpha_q-\hat{\gamma}a_q^2)\omega_qa_q+4\hat{\mu}_q\omega_ya_y\cos\phi_y)a_p^2a_q\\
m_{10}&=-\omega_q^2(1+\omega_q^2)((4\alpha_q-\hat{\gamma}a_q^2)\omega_qa_q+4\hat{\mu}_q\omega_ya_y\cos\phi_y)^2a_q^2\\
m_{21}&=\omega_y\omega_p\omega_q^2(4\hat{\alpha}_p-\hat{\gamma}a_p^2)a_p^2a_y\\
m_{22}&=\omega_y\omega_p^3(4\hat{\alpha}_p-\hat{\gamma}a_p^2)a_p^2a_y
\end{aligned}$$

$$
\begin{aligned}
m_{23} &= \omega_y\omega_p^3\omega_q^2(4\hat{\alpha}_p-\hat{\gamma}a_p^2)a_p^2a_y\\
m_{24} &= -2\omega_y\omega_p^2\omega_q(4\hat{\alpha}_p-\hat{\gamma}a_p^2)a_p^2a_y\\
m_{27} &= \omega_q^2a_q(4(\hat{\eta}u^2\omega_q^2a_q^2+\hat{\mu}_q\omega_y^2a_y^2)\cos\phi_y\\
&\quad -4\hat{\mu}_q\omega_q\omega_y^2a_y^2\sin\phi_y+(4\alpha_q-\hat{\gamma}a_q^2)\omega_y\omega_qa_ya_q)\\
m_{28} &= -\omega_p\omega_qa_q(4(\hat{\eta}u^2\omega_q^2a_q^2+\hat{\mu}_q\omega_y^2a_y^2)\cos\phi_y\\
&\quad -4\hat{\mu}_q\omega_q\omega_y^2a_y^2\sin\phi_y+(4\alpha_q-\hat{\gamma}a_q^2)\omega_y\omega_qa_ya_q)\\
m_{29} &= \omega_p\omega_q^2a_q(4\omega_q(\hat{\eta}u^2a_q^2-\hat{\mu}_q\omega_y^2a_y^2)\cos\phi_y\\
&\quad -4\hat{\mu}_q\omega_y^2a_y^2\sin\phi_y-(4\alpha_q-\hat{\gamma}a_q^2)\omega_y\omega_q^2a_ya_q)\\
m_{31} &= 2\omega_p^2\omega_q^2(4\hat{\alpha}_p-\hat{\gamma}a_p^2)a_p^4\\
m_{32} &= 2\omega_p^4(4\hat{\alpha}_p-\hat{\gamma}a_p^2)a_p^4\\
m_{33} &= 2\omega_p^4\omega_q^2(4\hat{\alpha}_p-\hat{\gamma}a_p^2)a_p^4\\
m_{34} &= -4\omega_p^3\omega_q(4\hat{\alpha}_p-\hat{\gamma}a_p^2)a_p^4\\
m_{37} &= \omega_p\omega_q^2a_p^2a_q(8\hat{\mu}_q\omega_ya_y(\cos\phi_y-\omega_q\sin\phi_y)\\
&\quad +(\hat{\gamma}a_p^2-4\hat{\alpha}_p-2\hat{\gamma}a_q^2+8\alpha_q)\omega_qa_q)\\
m_{38} &= -\omega_p^2\omega_qa_p^2a_q(8\hat{\mu}_q\omega_ya_y(\cos\phi_y-\omega_q\sin\phi_y)\\
&\quad +(\hat{\gamma}a_p^2-4\hat{\alpha}_p-2\hat{\gamma}a_q^2+8\alpha_q)\omega_qa_q)\\
m_{39} &= -\omega_p^2\omega_q^2a_p^2a_q(8\hat{\mu}_q\omega_ya_y(\omega_q\cos\phi_y+\sin\phi_y)\\
&\quad +(\hat{\gamma}a_p^2-4\hat{\alpha}_p-2\hat{\gamma}a_q^2+8\alpha_q)\omega_q^2a_q)\\
m_{30} &= -\omega_q^3(1+\omega_q^2)((4\alpha_q-\hat{\gamma}a_q^2)\omega_qa_q+4\hat{\mu}_q\omega_ya_y\cos\phi_y)a_q^3,
\end{aligned}
\tag{62}
$$

where $\sin\phi_y=\omega_y\hat{\delta}_1a_y/\hat{\eta}u^2a_q$ and $\cos\phi_y=\pm\sqrt{1-(\omega_y\delta_1a_y/\eta u^2a_q)^2}$.

Appendix E. Asymptotic equations for the 1:1:5:4:2:ω_r internal resonance

Substitution of (52) into (51) yields the following set of non-homogeneous equations:

$$
\begin{aligned}
D_0^2s_2+\omega_s^2s_2 &= -2i\omega_sD_1Ae^{i\omega_sT_0}-i\hat{\delta}_1(\omega_pH_sPe^{i\omega_pT_0}+\omega_sAe^{i\omega_sT_0})\\
&\quad -\hat{k}_{w1}(Ce^{i\omega_wT_0}+H_wRe^{i\omega_rT_0})\\
&\quad -k_{ss1}(A^2e^{2i\omega_sT_0}+|A|^2+H_s^2P^2e^{2i\omega_pT_0}+H_s^2|P|^2\\
&\quad +2AH_sPe^{i(\omega_s+\omega_p)T_0}+2AH_s\overline{P}e^{i(\omega_s-\omega_p)T_0})\\
&\quad -k_{yy1}(B^2e^{2i\omega_yT_0}+|B|^2+H_y^2Q^2e^{2i\omega_qT_0}+H_y^2|Q|^2\\
&\quad +2BH_yQe^{i(\omega_y+\omega_q)T_0}+2\overline{B}H_yQe^{i(\omega_q-\omega_y)T_0})\\
&\quad -k_{ww1}(C^2e^{2i\omega_wT_0}+|C|^2+H_w^2R^2e^{2i\omega_rT_0}+H_w^2|R|^2\\
&\quad +2CH_wRe^{i(\omega_w+\omega_r)T_0}+2CH_w\overline{R}e^{i(\omega_w-\omega_r)T_0})\\
&\quad -k_{sw1}(A\overline{C}e^{i(\omega_s-\omega_w)T_0}+ACe^{i(\omega_s+\omega_w)T_0}\\
&\quad +CH_s\overline{P}e^{i(\omega_w-\omega_p)T_0}+CH_sPe^{i(\omega_w+\omega_p)T_0}\\
&\quad +\overline{A}H_wRe^{i(\omega_r-\omega_s)T_0}+AH_wRe^{i(\omega_s+\omega_r)T_0}+H_sPH_w\overline{R}e^{i(\omega_p-\omega_r)T_0}\\
&\quad +H_sPH_wRe^{i(\omega_p+\omega_r)T_0})+\text{cc}\\
D_0^2y_2+\omega_y^2y_2 &= -2i\omega_yD_1Be^{i\omega_yT_0}-i\hat{\delta}_1(\omega_qH_yQe^{i\omega_qT_0}+\omega_yBe^{i\omega_yT_0})\\
&\quad -k_{sy2}(ABe^{i(\omega_s+\omega_y)T_0}+A\overline{B}e^{i(\omega_s-\omega_y)T_0}\\
&\quad +BH_sPe^{i(\omega_y+\omega_p)T_0}+BH_s\overline{P}e^{i(\omega_y-\omega_p)T_0}\\
&\quad +AH_yQe^{i(\omega_s+\omega_q)T_0}+\overline{A}H_yQe^{i(\omega_q-\omega_s)T_0}\\
&\quad +H_sPH_yQe^{i(\omega_p+\omega_q)T_0}+H_sPH_y\overline{Q}e^{i(\omega_p-\omega_q)T_0})\\
&\quad -k_{yw2}(BCe^{i(\omega_y+\omega_w)T_0}+B\overline{C}e^{i(\omega_y-\omega_w)T_0}\\
&\quad +BH_wRe^{i(\omega_y+\omega_r)T_0}+\overline{B}H_wRe^{i(\omega_r-\omega_y)T_0}\\
&\quad +CH_yQe^{i(\omega_w+\omega_q)T_0}+CH_y\overline{Q}e^{i(\omega_w-\omega_q)T_0}\\
&\quad +H_yQH_wRe^{i(\omega_q+\omega_r)T_0}+H_yQH_w\overline{R}e^{i(\omega_q-\omega_r)T_0})+\text{cc}\\
D_0^2w_2+\omega_w^2w_2 &= -2i\omega_wD_1Ce^{i\omega_wT_0}-i\hat{\delta}_1(\omega_rH_wRe^{i\omega_rT_0}+\omega_wCe^{i\omega_wT_0})\\
&\quad -\hat{k}_{s3}(Ae^{i\omega_sT_0}+H_sPe^{i\omega_pT_0})
\end{aligned}
$$

$$
\begin{aligned}
&-k_{ss3}(A^2e^{2i\omega_sT_0}+|A|^2+H_s^2P^2e^{2i\omega_pT_0}+H_s^2|P|^2\\
&+2AH_sPe^{i(\omega_s+\omega_p)T_0}+2AH_s\overline{P}e^{i(\omega_s-\omega_p)T_0})\\
&-k_{yy3}(B^2e^{2i\omega_yT_0}+|B|^2+H_y^2Q^2e^{2i\omega_qT_0}+H_y^2|Q|^2\\
&+2BH_yQe^{i(\omega_y+\omega_q)T_0}+2\overline{B}H_yQe^{i(\omega_q-\omega_y)T_0})\\
&-k_{ww3}(C^2e^{2i\omega_wT_0}+|C|^2+H_w^2R^2e^{2i\omega_rT_0}+H_w^2|R|^2\\
&+2CH_wRe^{i(\omega_w+\omega_r)T_0}+2CH_w\overline{R}e^{i(\omega_w-\omega_r)T_0})\\
&-k_{sw3}(A\overline{C}e^{i(\omega_s-\omega_w)T_0}+ACe^{i(\omega_s+\omega_w)T_0}\\
&+CH_s\overline{P}e^{i(\omega_w-\omega_p)T_0}+CH_sPe^{i(\omega_w+\omega_p)T_0}\\
&+\overline{A}H_wRe^{i(\omega_r-\omega_s)T_0}+AH_wRe^{i(\omega_s+\omega_r)T_0}\\
&+H_sPH_w\overline{R}e^{i(\omega_p-\omega_r)T_0}+H_sPH_wRe^{i(\omega_p+\omega_r)T_0})+\text{cc}\\
D_0^2p_2+\omega_p^2p_2=&-2i\omega_pD_1Pe^{i\omega_pT_0}-i\omega_p\hat{\gamma}_pP^3e^{3i\omega_pT_0}\\
&+i\omega_p(\hat{\alpha}_p-\hat{\gamma}_p|P|^2)Pe^{i\omega_pT_0}+i\hat{\mu}_p(\omega_pH_sPe^{i\omega_pT_0}+\omega_sAe^{i\omega_sT_0})\\
&-i\omega_qk_{p2}Q^2e^{2i\omega_qT_0}+\omega_q^2k_{p3}(Q^2e^{2i\omega_qT_0}-|Q|^2)+\text{cc}\\
D_0^2q_2+\omega_q^2q_2=&-2i\omega_qD_1Qe^{i\omega_qT_0}-i\omega_q\hat{\gamma}_qQ^3e^{3i\omega_qT_0}+i\omega_q(\hat{\alpha}_q-\hat{\gamma}_q|Q|^2)Qe^{i\omega_qT_0}\\
&+i\hat{\mu}_q(\omega_qH_yQe^{i\omega_qT_0}+\omega_yBe^{i\omega_yT_0})\\
&-i\omega_qk_{q2}(PQe^{i(\omega_p+\omega_q)T_0}-P\overline{Q}e^{i(\omega_p-\omega_q)T_0})\\
&-i\omega_pk_{q3}(PQe^{i(\omega_p+\omega_q)T_0}+P\overline{Q}e^{i(\omega_p-\omega_q)T_0})\\
&+\omega_p\omega_qk_{q4}(PQe^{i(\omega_p+\omega_q)T_0}-P\overline{Q}e^{i(\omega_p-\omega_q)T_0})\\
&-i\omega_qk_{q6}(QRe^{i(\omega_q+\omega_r)T_0}+Q\overline{R}e^{i(\omega_q-\omega_r)T_0})\\
&-i\omega_rk_{q7}(QRe^{i(\omega_q+\omega_r)T_0}-Q\overline{R}e^{i(\omega_q-\omega_r)T_0})\\
&+\omega_q\omega_rk_{q8}(QRe^{i(\omega_q+\omega_r)T_0}-Q\overline{R}e^{i(\omega_q-\omega_r)T_0})+\text{cc}\\
D_0^2r_2+\omega_r^2r_2=&-2i\omega_rD_1Re^{i\omega_rT_0}-i\omega_r\hat{\gamma}_rR^3e^{3i\omega_rT_0}+i\omega_r(\hat{\alpha}_r-\hat{\gamma}_r|R|^2)Re^{i\omega_rT_0}\\
&+i\hat{\mu}_r(\omega_rH_wRe^{i\omega_rT_0}+\omega_rCe^{i\omega_wT_0})-i\omega_qk_{r2}Q^2e^{2i\omega_qT_0}\\
&-\omega_q^2k_{r3}(Q^2e^{2i\omega_qT_0}-|Q|^2)+\text{cc}
\end{aligned}
\tag{63}
$$

Appendix F. Evolution equations and limit-cycle amplitude equations for the 1:1:5:4:2:ω_r internal resonance

The ten evolution equations in the 1:1:5:4:2:ω_r case are

$$
\begin{aligned}
&\omega_sD_1a_s=-\frac{1}{2}\hat{\delta}_1\omega_sa_s-\frac{1}{2}k_{yy1}H_ya_ya_q\sin\phi_1-\frac{1}{4}k_{sw1}H_sa_wa_p\sin\phi_2\\
&\omega_sa_sD_1\theta_s=\frac{1}{2}k_{yy1}H_ya_ya_q\cos\phi_1+\frac{1}{4}k_{sw1}H_sa_wa_p\cos\phi_2\\
&\omega_yD_1a_y=-\frac{1}{2}\hat{\delta}_1\omega_ya_y-\frac{1}{4}k_{sy2}H_ya_sa_q\sin\phi_1\\
&\omega_ya_yD_1\theta_y=\frac{1}{4}k_{sy2}H_ya_sa_q\cos\phi_1\\
&\omega_wD_1a_w=-\frac{1}{2}\hat{\delta}_1\omega_wa_w+\frac{1}{2}k_{ss3}H_sa_sa_p\sin\phi_2\\
&\omega_wa_wD_1\theta_w=\frac{1}{2}k_{ss3}H_sa_sa_p\cos\phi_2\\
&\omega_pD_1a_p=\frac{1}{2}\hat{\alpha}_p\omega_pa_p-\frac{1}{8}\hat{\gamma}_p\omega_pa_p^3+\frac{1}{2}\hat{\mu}_pH_s\omega_pa_p-\frac{1}{4}k_{p2}\omega_qa_q^2\cos\phi_p+\frac{1}{4}k_{p3}\omega_q^2a_q^2\sin\phi_p\\
&\omega_pa_pD_1\theta_p=-\frac{1}{4}k_{p2}\omega_qa_q^2\sin\phi_p-\frac{1}{4}k_{p3}\omega_q^2a_q^2\cos\phi_p\\
&\omega_qD_1a_q=\frac{1}{2}\hat{\alpha}_q\omega_qa_q-\frac{1}{8}\hat{\gamma}_q\omega_qa_q^3+\frac{1}{2}\hat{\mu}_qH_y\omega_qa_q\\
&\qquad+\frac{1}{4}k_{q2}\omega_qa_pa_q\cos\phi_p-\frac{1}{4}k_{q3}\omega_pa_pa_q\cos\phi_p+\frac{1}{4}k_{q4}\omega_p\omega_qa_pa_q\sin\phi_p\\
&\omega_qa_qD_1\theta_q=-\frac{1}{4}k_{q2}\omega_qa_pa_q\sin\phi_p+\frac{1}{4}k_{q3}\omega_pa_pa_q\sin\phi_p+\frac{1}{4}k_{q4}\omega_p\omega_qa_pa_q\cos\phi_p,
\end{aligned}
\tag{64}
$$

where the phases are $\phi_1=\theta_q-\theta_s-\theta_y+\sigma_1 T_1$ and $\phi_2=\theta_w-\theta_p-\theta_s+\sigma_2 T_1$. By setting the time derivatives of the final form of the evolution equations to zero, we have the eight coupled algebraic equations:

$$
\begin{aligned}
&2\hat{\delta}_1\omega_s a_s+2k_{yy1}H_y a_y a_q\sin\phi_1+k_{sw1}H_s a_w a_p\sin\phi_2=0\\
&2\hat{\delta}_1\omega_y a_y+k_{sy2}H_y a_s a_q\sin\phi_1=0\\
&\hat{\delta}_1\omega_w a_w+k_{ss3}H_s a_s a_p\sin\phi_2=0\\
&4\hat{\alpha}_p\omega_p a_p-\hat{\gamma}_p\omega_p a_p^3+4\hat{\mu}_p H_s\omega_p a_p-2k_{p2}\omega_q a_q^2\cos\phi_p+2k_{p3}\omega_q^2 a_q^2\sin\phi_p=0\\
&4\hat{\alpha}_q\omega_q a_q-\hat{\gamma}_q\omega_q a_q^3+4\hat{\mu}_q H_y\omega_q a_q+2k_{q2}\omega_q a_p a_q\cos\phi_p-2k_{q3}\omega_p a_p a_q\cos\phi_p\\
&\quad+2k_{q4}\omega_p\omega_q a_p a_q\sin\phi_p=0\\
&k_{q2}\omega_s\omega_y\omega_q a_s a_y a_p\sin\phi_p-k_{q3}\omega_s\omega_y\omega_p a_s a_y a_p\sin\phi_p-k_{q4}\omega_s\omega_y\omega_p\omega_q a_s a_y a_p\cos\phi_p\\
&\quad+2k_{yy1}H_y\omega_y\omega_q a_y^2 a_q\cos\phi_1+k_{sw1}H_s\omega_y\omega_q a_y a_w a_p\cos\phi_2\\
&\quad+k_{sy2}H_y\omega_s\omega_q a_s^2 a_q\cos\phi_1-4\sigma_1\omega_s\omega_y\omega_q a_s a_y=0\\
&2k_{ss3}H_s\omega_s\omega_p a_s^2 a_p^2\cos\phi_2+k_{p2}\omega_s\omega_w\omega_q a_s a_w a_q^2\sin\phi_p+k_{p3}\omega_s\omega_w\omega_q^2 a_s a_w a_q^2\cos\phi_p\\
&\quad-2k_{yy1}H_y\omega_w\omega_p a_y a_w a_p a_q\cos\phi_1-k_{sw1}H_s\omega_w\omega_p a_w^2 a_p^2\cos\phi_2\\
&\quad+4\sigma_2\omega_s\omega_w\omega_p a_s a_w a_p=0\\
&2k_{q2}\omega_p\omega_q a_p^2\sin\phi_p-2k_{q3}\omega_p^2 a_p^2\sin\phi_p-2k_{q4}\omega_p^2\omega_q a_p^2\cos\phi_p-k_{p2}\omega_q^2 a_q^2\sin\phi_p\\
&\quad+k_{p3}\omega_q^3 a_q^2\cos\phi_p-4\sigma_p\omega_p\omega_q a_p=0.
\end{aligned}
\tag{65a–h}
$$

We can eliminate ϕ_1 and ϕ_2 by isolating $\sin\phi_1$ and $\sin\phi_2$ in (65a–hb) and (65a–hc). Also we can eliminate ϕ_p by solving (65a–hd) and (65a–he) for $\sin\phi_p$ and $\cos\phi_p$. After elimination of the phases, we obtain five equations with five unknown state variables a_s, a_y, a_w, a_p, a_q. Like before we make use of the measured transverse limit-cycle amplitude in GW 97 experiment to obtain a_y. Note that Govardhan and Williamson (2005) also documented the streamwise limit-cycle amplitude. After $U^*=8$, the ratio between the transverse and streamwise amplitude becomes nearly constant. This information is utilized to determine a_s. The estimation of the rest of the state variables comes from previous numerical simulation. Specifically, we get the ratios between the desired variable and a_y at $U^*=20$. We assume $\gamma_p=\gamma_q=\gamma$, $\sigma_q=\sigma_2=\sigma_p=0$, $k_{p2}=k_{p3}=1$ and $k_{q2}=-2k_{q3}$, to minimize the number of unknowns. The last assumption comes from the previous observation in the 1:1:5:2:1:1 case.

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